
A Separation Principle for Cooperative Multi-Agent Reinforcement Learning

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Abstract

1 We study cooperative multi-agent reinforcement learning (MARL) problems in
2 which agents evolve under decoupled noisy dynamics but are coupled via a popula-
3 tion objective. A canonical example is a mobility operator that orchestrates a fleet
4 of vehicles in an urban environment to meet customer demand. The stochastic and
5 time-varying nature of these problems calls for learning approaches, yet existing
6 MARL algorithms do not scale: learning assignment and routing jointly over the
7 full state-action space becomes prohibitive as the number of agents grows. By
8 studying the problem as a control problem over probability measures, we prove a
9 separation principle: the population-level cost-to-go function is upper-bounded by
10 an optimal transport problem whose transportation cost is the target-conditioned
11 single-agent cost-to-go function. This result simplifies the multi-agent problem to
12 learning a single-agent policy and coordinating the agents via optimal transport.
13 Thus, we introduce SALT (Separation-based Assignment and Learning via optimal
14 Transport) a family of algorithms that learn a target-conditioned single-agent pol-
15 icy using any standard RL algorithm, and then coordinate the agents via optimal
16 transport. Because learning occurs at the single-agent level, SALT scales to fleet
17 sizes where MARL baselines become intractable and transfers without retraining
18 to fleet sizes unseen during training. In a vehicle-delivery experiment in south
19 Manhattan’s road network, where MARL baselines are computationally prohibitive,
20 SALT improves travel times by up to 20% over a shortest-path routing heuristic
21 and transfers without retraining to different fleet sizes. In stochastic grid worlds,
22 SALT reduces average travel times by more than 40% relative to MARL baselines
23 and scales to population sizes where these methods fail.

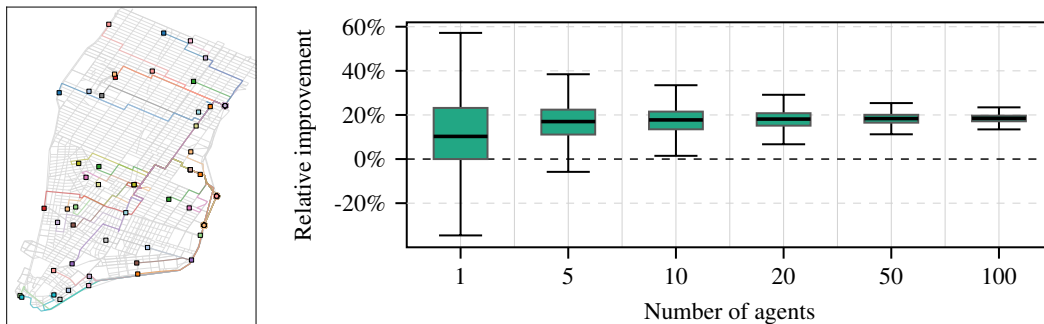


Figure 1: Relative improvement in travel time of SALT over the shortest-path baseline on the time-varying Manhattan road network, as a function of fleet size. Boxplots show the distribution across 500 evaluation instances and 3 seeds, and the dashed line indicates zero improvement. All models are trained with 20 vehicles and evaluated without retraining on fleets of 1-100 vehicles.

24 1 Introduction

25 Consider a fleet of autonomous vehicles serving delivery requests in a city. At each decision
 26 time, the operator must simultaneously decide which vehicle should serve which request and how
 27 each vehicle should navigate to its assigned destination. These decisions are, in principle, tightly
 28 coupled. Assignments based only on nominal shortest paths can be poor when travel times depend
 29 on traffic lights, congestion, and random delays. The best assignment depends on how long it will
 30 actually take each vehicle to reach each target, which depends on the routing, and hence on the
 31 assignment itself. Yet, solving the full joint assignment-and-routing problem over the fleet quickly
 32 becomes computationally prohibitive. Beyond delivery fleets, the same structure arises across a wide
 33 range of cooperative systems: inspection drones achieving coverage patterns [1], warehouse robots
 34 repositioning to incoming orders [2], emergency response units relocating to optimal positions [3],
 35 ride-hailing platforms balancing supply and demand [4], and electric vehicle sharing charging
 36 infrastructure [5]. In all these settings, the goal is to optimally coordinate a fleet of decoupled agents
 37 so that its collective terminal state matches or closely approximates a desired target configuration: a
 38 spatially uniform coverage, a set of delivery locations, or a demand-weighted positioning pattern.
 39 These problems are typically solved in a receding-horizon fashion to account for changing conditions
 40 and request patterns and, thus, they require modular approaches that generalize without retraining
 41 when the fleet size or target distribution change.

42 A standard approach for these problems is multi-agent reinforcement learning (MARL) [6, 7, 8, 9],
 43 often under centralized training with decentralized execution. Representative approaches include
 44 policy-gradient methods such as MAPPO [10] and MADDPG [11] and value-decomposition methods
 45 such as VDN [12], QMIX [13], and QPLEX [14]. Although effective in many settings [15, 16], theof
 46 their learning complexity therefore grows prohibitively with the number of agents [17], limiting
 47 the scalability and generalization to fleet sizes not seen during training. Value-decomposition
 48 methods gain tractability by imposing architectural assumptions on the joint value function, such
 49 as additivity or monotonicity, which however hold only in very specific settings [18]. Mean-field
 50 reinforcement learning [19, 20] reduces complexity by replacing pairwise interactions with an
 51 aggregate representation of the population. Despite being practical in large systems [4, 21], mean-
 52 field approaches rely on approximations that can lose fidelity when modeling finite populations [22].
 53 A separate line of work addresses assignment via combinatorial solvers or attention-based policies
 54 [23, 24], but treats routing as a downstream heuristic step. Such methods fail to take into account
 55 that a correct assignment cost is itself a stochastic control quantity. Heuristics can therefore assign
 56 vehicles to targets that “look close” but are expensive in expectation.

57 In this work, we seek a principled way to separate the population-level assignment and the single-agent
 58 control problem. Inspired by the “deterministic” separation principle for multi-agent control in [25],
 59 we cast the optimal control problem for the population as an optimal control problem over the space
 60 of probability measures where the control costs are optimal transport costs (e.g., Wasserstein distance)
 61 to a desired target measure. By construction, this representation is invariant to the permutations and
 62 can capture both finite populations (as empirical measures) and a continuum of agents (as continuous
 63 measures). In the deterministic setting, [25] shows that the dynamic programming recursion over the
 64 space of probability measures can be separated into an optimal transport problem and a single-agent
 65 control problem. However, this result does not extend exactly to the stochastic setting. Nevertheless,
 66 we show here that a carefully designed separation principle, where noise is “averaged out”, continues
 67 to upper bound the control cost. This analysis is non-trivial: In the stochastic setting, the probability
 68 measure capturing the population is itself random, yielding a probability measures over probability
 69 measures—a modeling challenge itself, compounded by delicate measurability issues. Finally,
 70 differently from [25], we do not limit ourselves to a theoretical analysis, but leverage our separation
 71 principle to design scalable algorithms for MARL.

72 **Contributions.** Our contributions are as follows:

- 73 1. We formulate cooperative MARL with homogeneous agents, decoupled stochastic dynamics,
 74 and terminal optimal transport formation costs as an optimal control problem over probability
 75 measures. This formulation removes artificial agent labels, applies to both finite populations and
 76 continuum of agents, is naturally invariant to permutations of the agents, and captures terminal
 77 formation or coverage costs through optimal transport.
- 78 2. We prove the separation principle for this optimal control problem (Theorem 3.1, Part I). The
 79 population-level cost-to-go function is upper-bounded by an optimal transport problem whose

80 transportation cost is the target-conditioned single-agent cost-to-go function. The resulting
 81 problem decouples the allocation problem (the transportation costs are fixed) from the routing
 82 problem (which can be solved independently from the fleet). In the deterministic setting, this upper
 83 bound becomes exact, recovering the deterministic separation principle from [25, Corollary 4.2].
 84 Importantly, our proof is constructive and directly yields an implementable policy.

- 85 3. This separation yields SALT, a family of MARL algorithms that learn a shared target-conditioned
 86 single-agent policy and coordinate agents via optimal transport. SALT is agnostic to the choice
 87 of the single-agent algorithm and generalizes to population sizes not seen during training.
- 88 4. We validate SALT on a time-varying south Manhattan road network with 2088 intersections and
 89 4262 directed roads. SALT trained with $N = 20$ vehicles transfers without retraining to fleets of
 90 1-100 vehicles and reduces travel time by up to 20% relative to a nominal shortest-path routing
 91 heuristic. We also use stochastic grid worlds for controlled ablations. SALT variants outperform
 92 standard MARL baselines by more than 40% in average travel time, consistently across different
 93 single-agent RL algorithms. Additionally, SALT greatly improves sample efficiency: with single-
 94 agent PPO, it needs up to $450\times$ fewer environment interactions than MAPPO and scales to
 95 $N = 128$, while MAPPO already fails at $N = 8$ within the 2×10^9 -interaction budget.

96 **Notation.** We work with Polish spaces, Borel functions, and Borel probability measures. For a
 97 Polish space X , $\mathcal{P}(X)$ denotes the space of probability measures on X . We write δ_x for the Dirac
 98 measure at $x \in X$. The projection from $X \times Y$ onto X is denoted as $\text{proj}_X^{X \times Y}$. For $\mu \in \mathcal{P}(X)$ and
 99 $T: X \rightarrow Y$, the pushforward of μ through T , $T_\# \mu \in \mathcal{P}(Y)$, is defined by $T_\# \mu(B) = \mu(T^{-1}(B))$
 100 for all Borel sets $B \subseteq Y$. A coupling between the probability measures $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ is
 101 a joint probability measure with marginals μ and ν ; the set of couplings between μ and ν is $\Gamma(\mu, \nu)$.
 102 For a nonnegative cost $c: X \times Y \rightarrow \mathbb{R}_{\geq 0}$, the optimal transport cost between μ and ν is

$$\mathcal{K}[c](\mu, \nu) := \inf_{\gamma \in \Gamma(\mu, \nu)} \int_{X \times Y} c(x, y) d\gamma(x, y).$$

103 If $k: X \rightarrow \mathcal{P}(Y)$ is a Markov kernel, $\mu \otimes k$ denotes the induced joint law on $X \times Y$. In particular,
 104 if $\pi(dx, dy) = \mu(dx)k(x; dy)$ is a disintegration, we write $\pi = \mu \otimes k^x$.

105 2 Problem setting

106 We consider a class of cooperative MARL problems in which N homogeneous agents coordinate to
 107 reach a desired configuration over a finite horizon T , while incurring agent- and terminal population-
 108 level costs. Let S , A , W , and Z denote the state, action, noise, and target spaces. While $Z = S$ in
 109 the canonical target matching, we keep Z distinct for generality: targets may be lower-dimensional
 110 or task-level descriptors, such as destinations when the state also includes velocity or battery level.

111 2.1 Agent-based formulation

112 Each agent i has state $s_t^i \in S$ and chooses action $a_t^i \in A$. Its noisy dynamics are

$$s_{t+1}^i = f_t(s_t^i, a_t^i, w_t^i), \quad w_t^i \sim \xi_t(\cdot | s_t^i, a_t^i),$$

113 where $\xi_t(\cdot | s_t^i, a_t^i) \in \mathcal{P}(W)$ gives the conditional law of the noise affecting an agent in state s
 114 taking action a . We consider three noise models, which differ in how noise is correlated across
 115 agents: *individual noise*; where noise is sampled independently across agents, *local noise*; where
 116 agents with the same state-action pair share the same noise realization, and *global noise*, where
 117 all agents share a common noise realization. Their formal definition can be found in Appendix A.
 118 Let $\mathbf{s}_t := (s_t^1, \dots, s_t^N)$ and $\mathbf{a}_t := (a_t^1, \dots, a_t^N)$ denote the ordered agents' states and actions.
 119 The control cost results from the sum of agent-level stage costs $g_t: S \times A \times W \rightarrow \mathbb{R}_{\geq 0}$ and a
 120 population-level terminal formation cost. The terminal cost compares the final population distribution
 121 $\mu_T = \frac{1}{N} \sum_{i=1}^N \delta_{s_T^i}$ to a target configuration $\nu \in \mathcal{P}(Z)$ through the optimal transport cost

$$\mathcal{K}[g_T](\mu_T, \nu) = \inf_{\gamma \in \Gamma(\mu_T, \nu)} \int_{S \times Z} g_T(s, z) d\gamma(s, z), \quad (1)$$

122 where $g_T: S \times Z \rightarrow \mathbb{R}_{\geq 0}$ is the transportation cost; e.g., when $g_T(s, z) = \|s - z\|^p$ for $p \geq 1$, then
 123 $\mathcal{K}[g_T](\mu_T, \nu)$ is the p power of the type- p Wasserstein distance. Since the problem is a fully observed

cooperative MARL problem, it can be cast as a Multiagent MDPs [26], which reduces to a standard single-agent MDP over the joint action space. By standard MDP theory [27], we losslessly consider only deterministic (measurable) Markov policies $\pi = (\pi_t)_{t=0}^{T-1}$. We denote this class by Π .

Problem 1. Given initial states $\bar{s}_0 = (\bar{s}_0^1, \dots, \bar{s}_0^N)$ and a target distribution ν , learn a policy $\pi \in \Pi$ minimizing the expected cumulative cost

$$J^N(\bar{s}_0, \nu) = \inf_{\pi \in \Pi} \mathbb{E}_{(w_t)_{t=0}^{T-1}} \left[\frac{1}{N} \sum_{i=1}^N \sum_{t=0}^{T-1} g_t(s_t^i, a_t^i, w_t^i) + \mathcal{K}[g_T](\mu_T, \nu) \right]$$

s.t. $s_{t+1}^i = f_t(s_t^i, a_t^i, w_t^i), \quad s_0 = \bar{s}_0.$

2.2 Population-based formulation

We generalize Problem 1 at the *population-level* by replacing the ordered state s_t with the empirical measure $\mu_t = \frac{1}{N} \sum_{i=1}^N \delta_{s_t^i} \in \mathcal{P}(S)$ which discards agent labels and makes the permutation invariance in the problem explicit. For now, we allow general population states $\mu_t \in \mathcal{P}(S)$; we will later specialize this formulation back to the finite-agent setting.

Population control. A population-level control must specify how the mass at each state is split among actions. Such a control is a *state-action distribution* $\lambda_t \equiv \lambda_t(\mu_t) \in \mathcal{P}(S \times A)$ whose S -marginal is the current population state: $(\text{proj}_S^{S \times A})_{\#} \lambda$, the pushforward via the projection map $\text{proj}_S^{S \times A}$. Let

$$M(\mu_t) := \{\lambda \in \mathcal{P}(S \times A) : (\text{proj}_S^{S \times A})_{\#} \lambda = \mu_t\}$$

denote the set of admissible state-action distributions at population state μ_t . For example, if all agents follow a common deterministic policy $\pi_t : S \rightarrow A$, then $\lambda_t = (\text{Id}_S, \pi_t)_{\#} \mu_t$ that is, each unit of mass at state s is placed at the state-action pair $(s, \pi_t(s))$.

Noise and population dynamics. When dynamics are deterministic, the next population state is the pushforward of μ under the dynamics, i.e., $\mu_{t+1} = f_t_{\#} \lambda_t$. The stochastic case, however, requires more care. The transition also depends on the specific noise realizations across the population. In the agent-based formulation, agent i simply draws $w_t^i \sim \xi_t(\cdot | s_t^i, a_t^i)$ and transitions to $f_t(s_t^i, a_t^i, w_t^i)$. At the population level, however, we cannot simply replace these individual draws by the underlying distribution $\xi_t(\cdot | s, a)$: the next population state is determined by the *realized collection* of noise draws across all agents, which is itself a random object. These draws are random with distribution conditioned over state-action pairs and hence need to be represented as a state-action dependent *random* probability measure $\hat{\xi}_t^{s,a} \in \mathcal{P}(W)$. The distinction between $\xi_t(\cdot | s, a)$ and $\hat{\xi}_t^{s,a}$ is crucial: the former is the single-agent noise law, while the latter collects the noise realizations that drive the population transition. Collecting the realized empirical kernels $\hat{\xi}_t^{s,a}$ across all state-action pairs, we define the joint state-action-noise distribution $\rho_t = \lambda_t \otimes \hat{\xi}_t^{s,a} \in \mathcal{P}(S \times A \times W)$. The distribution ρ_t records exactly the fraction of the population experiencing each triple (s, a, w) . The population then evolves according to the pushforward of the joint state-action-noise distribution ρ_t under the dynamics f_t , so that $\mu_{t+1} = f_t_{\#} \rho_t$.

Example 2.1. Consider three agents with the state-action distribution $\lambda_t = \frac{2}{3}\delta_{(0,a)} + \frac{1}{3}\delta_{(1,a)}$, i.e., two agents have state 0 and use action a , one agent has state 1 and uses action a . Assume that agents are subject to individual noise and that the noise realizations for the two agents at $(0, a)$ are 0 and 1, while the noise realization for the agent at $(1, a)$ is 1. Then, $\hat{\xi}_t^{0,a} = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ and $\hat{\xi}_t^{1,a} = \delta_1$. Accordingly, the state-action-noise distribution is $\rho_t := \lambda_t \otimes \hat{\xi}_t^{s,a} = \frac{1}{3}\delta_{(0,a,0)} + \frac{1}{3}\delta_{(0,a,1)} + \frac{1}{3}\delta_{(1,a,1)}$ and the next population distribution is $\mu_{t+1} = (f_t)_{\#} \rho_t = \frac{1}{3}\delta_{f_t(0,a,0)} + \frac{1}{3}\delta_{f_t(0,a,1)} + \frac{1}{3}\delta_{f_t(1,a,1)}$.

Cost. As $\frac{1}{N} \sum_{i=1}^N g_t(s_t^i, a_t^i, w_t^i) = \int_{S \times A \times W} g_t(s, a, w) d\rho_t(s, a, w)$, the control cost is

$$J(\bar{\mu}_0, \nu) = \mathbb{E}_{(\hat{\xi}_t)_{t=0}^{T-1}} \left[\sum_{t=0}^{T-1} \int_{S \times A \times W} g_t(s, a, w) d\rho_t(s, a, w) + \mathcal{K}[g_T](\mu_T, \nu) \right],$$

where $\hat{\xi}_t = (\hat{\xi}_t^{s,a})_{(s,a) \in S \times A}$ collects all realized noise kernels at time t . The expectation is then taken over the randomness of these profiles. This keeps finite-population fluctuations and noise correlations

explicit while reducing the population-level randomness to the single object $\hat{\xi}_t$. We refer the reader to Appendix A for additional details and explicit construction in canonical noise models.

Problem 2. Given an initial population distribution $\bar{\mu}_0$, find a sequence of state-action distributions $(\lambda_t)_{t=0}^{T-1}$ that minimizes the expected cumulative cost

$$J(\bar{\mu}_0, \nu) = \inf_{(\lambda_t)_{t=0}^{T-1}} \mathbb{E}_{(\hat{\xi}_t)_{t=0}^{T-1}} \left[\sum_{t=0}^{T-1} \int_{S \times A \times W} g_t(s, a, w) d\rho_t(s, a, w) + \mathcal{K}[g_T](\mu_T, \nu) \right]$$

s.t. $\mu_0 = \bar{\mu}_0, \quad \mu_{t+1} = f_{t\#}\rho_t, \quad (\text{proj}_S^{S \times A})_{\#}\lambda_t = \mu_t \quad \rho_t = \lambda_t \otimes \hat{\xi}_t^{s,a}.$

The population formulation removes the artificial labels from the state description and generalizes Problem 1. Any finite-agent policy induces an empirical state-action distribution $\lambda_t = \frac{1}{N} \sum_{i=1}^N \delta_{(s_t^i, a_t^i)} \in M(\mu_t)$, so every policy admissible for Problem 1 is feasible in Problem 2 and $J(\bar{\mu}_0, \nu) \leq J^N(\bar{s}_0, \nu)$. The converse inequality does not hold in general. Problem 2 allows for arbitrary population state-action distributions, which may split mass in ways that are not realizable by N agents. However, we show in Corollary 3.3 that the separation bound derived for Problem 2 directly upper bounds the agent-based cost to go $J^N(\bar{s}_0, \nu)$, yielding a deployable policy for Problem 1.

3 Separation principle

In this section, we prove a separation principle for Problem 2, showing that the population cost-to-go function separates into optimal single-agent control toward a target and optimal target assignment across the population; see Theorem 3.1 below. We then specialize the result for the finite-agent value J^N of Problem 1 and obtain a deployable policy that we will later use for algorithm design.

Target-conditioned single-agent problem. Fix a target $z \in Z$. For an agent in state $s \in S$ at time t , define $j_t(s, z)$ as the optimal cost to go to reach z over the remaining horizon:

$$j_t(s, z) = \inf_{(\pi_\tau)_{\tau=t}^{T-1}} \mathbb{E}_{(w_\tau)_{\tau=t}^{T-1}} \left[\sum_{\tau=t}^{T-1} g_\tau(s_\tau, a_\tau, w_\tau) + g_T(s_T, z) \right], \quad (2)$$

s.t. $s_t = s, \quad s_{\tau+1} = f_\tau(s_\tau, a_\tau, w_\tau), \quad w_\tau \sim \xi_\tau(\cdot | s_\tau, a_\tau).$

This is a standard single-agent stochastic optimal control problem—whose complexity is independent of the population size N —that can be studied via the dynamic programming principle:

$$j_t(s_t, z) = \inf_{a_t \in A} \mathbb{E}_{w_t} [g_t(s_t, a_t, w_t) + j_{t+1}(f_t(s_t, a_t, w_t), z)], \quad j_T(s_T, z) = g_T(s_T, z). \quad (3)$$

A separation principle. For $\mu_t \in \mathcal{P}(S)$, let $J_t(\mu, \nu)$ be the optimal expected cost from time t to the horizon, when the population starts from μ_t and aims to match the target distribution ν . At $t = T$, $J_T(\mu, \nu) = \mathcal{K}[g_T](\mu, \nu)$. Fix $t < T$. Since $\mathcal{P}(S)$ is a Polish space whenever S is Polish [28, Theorem 15.15], the cost is additive over time, and the population dynamics are Markovian, standard dynamic programming yields the population-level Bellman recursion [29, Ch. 8]

$$J_t(\mu_t, \nu) = \inf_{\lambda_t \in M(\mu_t)} \mathbb{E}_{\hat{\xi}_t} \left[\int_{S \times A \times W} g_t(s, a, w) d\rho_t(s, a, w) + J_{t+1}(f_{t\#}\rho_t, \nu) \right]. \quad (4)$$

We now state the main result of this work.

Theorem 3.1 (Separation principle, Part I: upper bound for the population cost to go). *Consider Problem 2. For all $t \in \{0, \dots, T\}$, the population cost-to-go function (4) is upper bounded by an optimal transport problem whose transportation cost is the single-agent cost-to-go function j_t in (2):*

$$J_t(\mu_t, \nu) \leq \mathcal{K}[j_t](\mu_t, \nu) = \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int_{S \times Z} j_t(s, z) d\gamma(s, z). \quad (5)$$

In particular, the inequality holds true for the cost to go $J(\bar{\mu}_0, \nu)$ in Problem 2.

Discussion. The upper bound (5) separates the stochastic population control problem into a single-agent control problem yielding a population-level matching problem in which j_t plays the role of the transport cost. This separation generally comes at the price of an inequality, which we can explain as follows. In the optimal transport cost (5), mass is assigned to targets based on $j_t(s, z)$, which *averages* over future noise realizations. By contrast, the Bellman recursion (4) evaluates the optimal cost-to-go function $J_{t+1}(f_{t\#}\rho_t, \nu)$ at the realized future population $f_{t\#}\rho_t$. Thus, the optimal transport coupling is here computed *after* observing the noise realization. As optimal transport costs entail a minimization, they generally do not commute with an expectation. Interchanging them is the cause of the inequality (5). While the inequality could, in principle, be improved by parameterizing the single-agent cost to go in the noise realizations, this would lead to an intractable single-agent problem. Instead, averaging out the noise leads to stochastic optimal control problems and, thus, appears to be the best practically relevant solution. When the dynamics are deterministic, the next population state is fully determined by (μ_t, λ_t) and the expectation in (4) drop. As such, our upper bound in Theorem 3.1, Part I becomes exact, recovering the deterministic separation of [25].

Corollary 3.2 (Exact separation in the deterministic setting [25, Corollary 4.2]). *When the dynamics are deterministic, the population cost-to-go function (4) is an optimal transport cost with the single-agent cost-to-go function as its transportation cost:*

$$J_t(\mu_t, \nu) = \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int_{S \times Z} j_t(s, z) d\gamma(s, z).$$

From a bound to a policy. The proof of Theorem 3.1, Part I constructs an explicit population-level control $(\lambda_t^\varepsilon)_{t=0}^{T-1}$ that achieves cost within any $\varepsilon > 0$ of the bound. We denote $J^\lambda(\bar{\mu}_0, \nu)$ the expected cumulative cost of Problem 2 when the population follows the state-action distributions $(\lambda_t)_{t=0}^{T-1}$.

Theorem 3.1 continued (Separation principle, Part II: constructive separation-based control). *For each timestep t and every $\varepsilon > 0$, let γ_t^ε be an $\frac{\varepsilon}{2T}$ -optimal solution of the transport problem $\mathcal{K}[j_t](\mu_t, \nu)$ and let $a_t^\varepsilon : S \times Z \rightarrow A$ be an $\frac{\varepsilon}{2T}$ -optimal single-agent action for the Bellman recursion (3). Define the separation-based state-action distribution, $\lambda_t^\varepsilon := (\text{proj}_S^{S \times Z}, a_t^\varepsilon)_\# \gamma_t^\varepsilon$. Then,*

$$J^{\lambda^\varepsilon}(\bar{\mu}_0, \nu) \leq \inf_{\gamma \in \Gamma(\mu_0, \nu)} \int_{S \times Z} j_0(s, z) d\gamma(s, z) + \varepsilon.$$

In particular, if both an optimal γ_t^ and optimal single-agent actions a_t^* exist (e.g., all functions are continuous and all spaces compact), we can choose $\varepsilon = 0$.*

This construction has a natural interpretation. Assuming, for simplicity, existence of optimal couplings and actions, the coupling $\gamma_t^* \in \Gamma(\mu_t, \nu)$ is the optimal assignment of population mass to targets: for each state s , it specifies how the mass at s is assigned to the targets. The state-action distribution then assigns to each (s, z) pair the single-agent optimal action a_t^* that steers the agent from s to z . This gives rise to our separation principle: the coupling γ_t^* indicates how to dispatch agents to targets, while a_t^* indicates how to steer the single agent to the assigned target. Theorem 3.1 rigorously quantifies the performance of this construction, which is optimal in a deterministic setting.

Implication on agent-based formulation J^N . Theorem 3.1 is stated for the population formulation in Problem 2. Corollary 3.3 bridges this to the original agent-based setting: for N agents and N target masses, the same transport surrogate $\mathcal{K}[j_t]$ upper bounds J_t^N .

Corollary 3.3 (Agent-based upper bound). *Let $\mathbf{s}_t = (s_t^1, \dots, s_t^N) \in S^N$ be the agent states at time t , with empirical distribution $\mu_t^N = \frac{1}{N} \sum_{i=1}^N \delta_{s_t^i}$. Let $\mathbf{z} = (z^1, \dots, z^N) \in Z^N$ be fixed targets with empirical distribution $\nu^N = \frac{1}{N} \sum_{i=1}^N \delta_{z^i}$. Then, the cost-to-go function for the optimal control problem in Problem 1, denoted by $J_t^N(\mathbf{s}_t, \nu^N)$, satisfies*

$$J_t^N(\mathbf{s}_t, \nu^N) \leq \mathcal{K}[j_t](\mu_t^N, \nu^N). \quad (6)$$

In particular, the inequality holds true for the cost to go $J^N(\bar{\mathbf{s}}_0, \nu^N)$ in Problem 1.

We defer the proof of Theorem 3.1, Part I, Theorem 3.1, Part II, and Corollary 3.3 to Appendix C.

Algorithm 1 SALT: Separation-based Assignment and Learning via optimal Transport

Require: Initial target-conditioned policy $\pi^{(0)}(a \mid s, z)$, cost-to-go function estimate $\hat{j}^{(0)}(s, z)$, horizon T , exploration schedule $(\alpha)_{k \geq 0}$

- 1: **for** episodes $k = 0, 1, 2, \dots$ **do**
- 2: Sample the episode target distribution ν_k .
- 3: **Population-level matching.** With probability $1 - \alpha_k$, assign agents to targets via optimal transport with cost $\hat{j}^{(k)}$. Otherwise assign randomly.
- 4: **Single-agent learning.** Roll out the shared target-conditioned policy $\pi^{(k)}$ in parallel for all agents over the horizon T , with the assigned targets kept fixed.
- 5: Update $\pi^{(k+1)}$ and $\hat{j}^{(k+1)}(s, z)$ with the chosen single-agent RL algorithm from the N trajectories collected in Step 4.
- 6: **end for**
- 7: **Evaluation:**
- 8: **for** $t = 0, \dots, T - 1$ **do**
- 9: Assign agents to targets via optimal transport with cost $\hat{j}_t$.
- 10: Execute one step of π_t with the current target assignments.
- 11: **end for**

214 4 SALT: Separation-based assignment and learning via optimal transport

215 Corollary 3.3 suggests using the upper bound

$$\mathcal{K}[j_t](\mu_t, \nu) = \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int j_t(s, z) d\gamma(s, z)$$

216 as a tractable surrogate for the agent-based cost to go $J_t^N(s_t, \nu^N)$. The resulting learning paradigm,
217 summarized in Algorithm 1, has two phases: (i) learn the single-agent target-conditioned cost-to-go
218 function $\hat{j}_t$, (ii) use it as the transport cost to assign agents to targets. The choice of single-agent
219 algorithm is entirely open—any method capable of learning a target-conditioned cost-to-go estimate
220 $\hat{j}_t$ (Q-learning [30], DDQN [31], PPO [32], A2C [33], etc.) can be plugged in as the base learner. We
221 refer to this family of methods as SALT, for Separation-based Assignment and Learning via optimal
222 Transport. An instance using a single-agent RL algorithm is denoted SALT-ALG; e.g., SALT-PPO.

223 **Discussion.** SALT replaces direct learning over the joint population state-action space with a target-
224 conditioned single-agent policy $\pi(\cdot \mid s, z)$ and cost-to-go estimate $\hat{j}(s, z)$ shared across all agents,
225 where the only agent-specific quantity is the target z assigned by the optimal transport step. This is
226 desirable for three main reasons. (i) Scalability: Learning occurs entirely at the target-conditioned
227 single-agent level, avoiding the curse of dimensionality of the joint state-action space and generalizing
228 naturally across population sizes. (ii) Trajectory pooling: The same single-agent model is updated
229 from every agent’s trajectory simultaneously. (iii) Modularity: The coordination layer requires only a
230 cost-to-go estimate $\hat{j}(s, z)$ and, thus, SALT can be paired with any single-agent RL algorithm.

231 **Training details.** The surrogate $\mathcal{K}[j_t](\mu_t, \nu)$ requires a cost-to-go estimate $\hat{j}_t(s, z)$ used as the cost
232 for population-level assignment, and a single-agent policy $\pi_t(a \mid s, z)$, executed once targets are
233 assigned. Both are functions only of a single agent’s state and target, so learning is entirely at the
234 single-agent level and independent of population size. At episode k , the initial distribution μ_0 and the
235 target distribution ν (which is part of task specification) are sampled and agents are assigned to targets
236 by solving the optimal transport problem using the current estimate $\hat{j}^{(k)}(s, z)$ as cost. For empirical
237 measures with N samples, this optimal transport problem reduces to an assignment problem. It can
238 be solved exactly in $O(N^3)$, or approximately using entropic regularization [34]. With probability
239 α_k assignments are drawn uniformly at random instead, ensuring broad coverage of (s, z) pairs and
240 preventing the cost-to-go estimate from overfitting to its own assignments. Assignments are kept
241 fixed for the entire episode. This makes each agent’s problem a standard single-agent task with
242 terminal cost $g_T(s_T, z^i)$, whose cost to go is exactly $j_t(s, z^i)$, and ensures that collected trajectories
243 are consistent with the target-conditioned objective. Agents then roll out the shared target-conditioned
244 policy and all transitions are pooled to update the shared policy/value model. Thus, a population of
245 N agents provides N rollouts per episode covering diverse (s, z) pairs simultaneously. The precise
246 update rule depends on the chosen single-agent RL algorithm. At deployment, the optimal transport

	Individual noise			Local noise			Global noise		
SALT-Q	24.2	29.4	39.5	24.2	29.4	39.6	24.2	29.0	38.9
SALT-DDQN	24.2	29.0	37.8	24.2	29.2	38.2	24.2	28.8	36.5
SALT-PPO	24.1	28.0	35.1	24.1	28.1	36.2	24.1	28.2	34.0
SALT-A2C	24.4	27.9	35.7	24.4	28.3	36.4	24.4	28.3	34.8
MAPPO	50.9	52.2	60.8	50.9	51.8	59.9	50.9	51.5	55.7
IPPO	47.4	52.5	59.8	47.4	50.1	59.0	47.4	50.2	54.6
QMIX	58.4	64.3	65.8	58.4	66.2	74.9	58.4	60.9	63.9
VDN	61.4	77.6	69.0	61.4	78.4	77.6	61.4	65.3	65.6
	0	0.10	0.25	0	0.10	0.25	0	0.10	0.25
	Noise probability			Noise probability			Noise probability		

Figure 2: Average travel time to targets as a function of the noise probability p under individual, local, and global noise. Entries are means across 20 evaluation seeds. Brighter cells indicate worse relative performance. SALT variants consistently achieve lower travel times than standard MARL baselines.

assignment is recomputed at each timestep with cost $\hat{j}_t(s, z)$, and each agent acts under its newly assigned target. This mirrors the separation bound in Theorem 3.1, Part I and allows the policy to correct for stochastic deviations that accumulate over the episode.

5 Experimental results

We evaluate SALT in stochastic grid worlds, where we compare against MARL baselines, and in an urban logistic task in Manhattan road network, where we test scalability and transfer across population sizes. Code is at anonymous.4open.science/r/SALT-17FB/. More ablations and experiment details are in Appendix D.

5.1 Stochastic grid-world experiments

Separation leads to better performance. We first evaluate SALT on a stochastic grid-world environment. A population of $N = 16$ homogeneous agents must reach N target locations within a horizon $T = 100$. Agents start from the first column of an 11×30 grid, targets are uniformly sampled in the last ten columns. At each step, the intended action is replaced by a random one with probability p . We evaluate the three noise models introduced in Section 2: *individual*, *local* and *global* noise. The objective is to minimize average travel time and terminal mismatch to the target distribution. We compare four instances of SALT, using Q-learning [35] (SALT-Q), DDQN [31] (SALT-DDQN), PPO [32] (SALT-PPO), and A2C [33] (SALT-A2C) as single-agent routing algorithms, against four common deep MARL baselines, MAPPO [10], IPPO [10], VDN [12], QMIX [13]. All methods are trained and evaluated under the same task distribution and random seeds. Figure 2 shows that SALT consistently outperforms the MARL baselines across all noise levels and noise models. The four SALT variants perform similarly, suggesting that the gain comes from the separation structure rather than from the single-agent RL algorithm. Complementary metrics are reported in Appendix D.

Separation leads to better scalability. We next study how sample complexity scales with the number of agents. We use the same grid-world dynamics with individual noise, but vary the number of agents from $N = 2$ to $N = 128$. We keep the target density roughly constant to isolate the effects of scaling from the complexity of the task and avoid scenarios with highly condensed targets. Agents start at the bottom-left corner, and targets are sampled uniformly across the grid (see Appendix D.2). We select PPO as the single-agent learner in SALT, yielding SALT-PPO, and compare against MAPPO. We measure the number of environment interactions required to reach reach-rate milestones of 70%, 80%, 90%, 95%, and 99%, up to a maximum training budget of $2e9$ interactions. Figure 3 shows that SALT-PPO scales substantially better than MAPPO. SALT-PPO reaches high reach rates across all tested population sizes, while MAPPO scales poorly and often fails to reach the same milestones within the training budget. This scalability gap is directly explained by the separation principle. MAPPO must learn in the joint multi-agent state-action space, whose dimensionality grows with N , whereas SALT-PPO learns a single target-conditioned policy that is independent of N .

		SALT-PPO (ours)								MAPPO			
Reach-rate milestone	99%	3.8e6	7.7e7	5.2e7	2.9e7	8.5e7	1.8e8	4.5e8	1.5e9	—	—	—	—
	95%	8.9e5	2.2e6	3.5e6	6.0e6	1.8e7	6.0e7	2.3e8	8.6e8	1.1e7	3.0e8	1.6e9	—
	90%	6.3e5	1.8e6	3.0e6	5.2e6	1.5e7	5.1e7	1.9e8	7.3e8	9.3e6	2.4e8	1.4e9	—
	80%	4.7e5	1.4e6	2.5e6	4.3e6	1.3e7	4.3e7	1.6e8	6.3e8	8.9e6	2.2e8	1.2e9	—
	70%	3.9e5	1.2e6	2.2e6	3.8e6	1.1e7	3.8e7	1.5e8	5.6e8	6.7e6	1.9e8	9.7e8	—
		$N=2$ 10×10	$N=4$ 14×14	$N=6$ 17×17	$N=8$ 20×20	$N=16$ 28×28	$N=32$ 40×40	$N=64$ 57×57	$N=128$ 80×80	$N=2$ 10×10	$N=4$ 14×14	$N=6$ 17×17	$N=8$ 20×20
		Agents / grid size								Agents / grid size			

Figure 3: Training environment interactions for a reach-rate milestone (70% – 99%) for SALT-PPO and MAPPO, as a function of population size N (grid size scaled proportionally to keep target density constant). A dash (–) indicates the milestone was not reached within the $2e9$ -interaction budget.

5.2 Towards a realistic time-varying stochastic urban logistics task

A fleet of delivery vehicles must serve requests distributed across Manhattan, jointly minimizing total travel time. This task couples vehicle-to-request assignment with time-efficient routing through a stochastic, time-varying network and is precisely the setting for which the separation principle was motivated. Rather than solving the full multi-agent problem online, SALT decomposes it into a coordination step (assigning vehicles to requests via optimal transport) and a routing step (navigating each vehicle to its target via a shared learned policy). The environment is a directed graph extracted from OpenStreetMap [36], containing the 2088 intersections and 4262 directed roads south of Central Park (see Figure 1). Travel times depend on road length, road type, traffic-light cycles of 90 seconds (with a road-dependent green phase duration and a random offset) and stochastic congestion effects. Further environment details are provided in Appendix D.3. A fleet of vehicles is initialized across three depot locations with random proportions and serves delivery requests sampled across the city, minimizing total travel time. We instantiate SALT with tabular Q-learning SALT-Q and compare it against an online vehicle-assignment baseline based on nominal shortest-path travel times. This baseline is a standard operational heuristic and ignores traffic-light timing and stochastic congestion, since these require spatiotemporal graphs whose size scales prohibitively with the horizon. The deep MARL baselines of Section 5.1 are excluded: as demonstrated, they become computationally prohibitive on a graph of this scale and are tied to a fixed training fleet size. All models are trained with $N = 20$ agents and evaluated, without retraining, on fleets of 1-100 vehicles.

Figure 1 reports the relative improvement in travel time as a function of fleet size. SALT-Q consistently outperforms the shortest-path baseline by roughly 20%, reflecting its ability to anticipate traffic-light timing and adapt to stochastic congestion. Performance remains stable across fleet sizes, confirming that SALT generalizes naturally to fleet sizes unseen at training time. The residual variance arises from a natural bias of distributional training: the policy concentrates on high-probability origin-destination pairs, improving sample efficiency at the cost of reduced accuracy on rare assignments.

6 Conclusion and limitations

We established a separation principle for cooperative MARL and used it to design SALT, a family of algorithms that learns a single-agent policy and uses optimal transport to coordinate agents. SALT outperforms existing MARL baselines in grid-worlds, both in terms of performance and scalability. In south Manhattan road network, where MARL baselines are impractical, SALT improves travel time by up to 20% over a shortest-path heuristic and generalizes to population sizes not seen during training.

Our work does not address several important challenges, which we leave to future work. From a theoretical perspective, our separation principle requires decoupled dynamics and an optimal transport terminal cost, ruling out settings with interactions, collisions, or non-transport objectives. Our bound on the optimal cost is also not tight in general. From a practical perspective, the target-conditioned single-agent problem must generalize across the target space, which is challenging when targets are high-dimensional or sampled from distributions far from the training one.

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A Noise models and averaging property

This appendix develops the measure-theoretic framework used to handle stochasticity in the multi-agent system. We first introduce the relevant general notation, and illustrate how common noise models fit into this framework through a series of examples. We conclude with Lemma A.9, which will be a key technical tool used in the proof of Theorem 3.1, Part I in Appendix C. Throughout, the space of probability measures on a Borel space X , $\mathcal{P}(X)$ is endowed with the topology of weak convergence and its Borel σ -algebra. Since X is a Polish space, $\mathcal{P}(X)$ is itself a Polish space [28, Theorem 15.15]. The product measure of two independent random variables with marginals $\mu \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Y)$ is the unique measure $\mu \otimes \nu \in \mathcal{P}(X \times Y)$ [37, Theorem B, Section 35] such that $(\mu \otimes \nu)(A \times B) = \mu(A) \nu(B)$ for all Borel sets $A \subseteq X$, $B \subseteq Y$.

A.1 General framework

In multi-agent systems, randomness enters at the *individual* level, through the noise realizations for each individual agent. At the *population* level, these realizations induce randomness in the population distribution and therefore in the next population state. To capture this in a unified way, we work with distributions over noise distributions. Concretely, following the notation in Section 2.2, let $\lambda_t \in \mathcal{P}(S \times A)$ denote the joint state-action distribution of the population at time t , and denote with $\xi_t(\cdot | s, a) \in \mathcal{P}(W)$ the noise kernel, i.e., the law of the noise experienced by an agent in state s taking action a . For agents sharing the same state-action pair (s, a) , their noise realizations form an empirical distribution that we denote $\hat{\xi}_t^{s,a} \in \mathcal{P}(W)$. Unlike the kernel $\xi_t(\cdot | s, a)$, which is a fixed object, $\hat{\xi}_t^{s,a}$ is random—it fluctuates across episodes depending on the specific noise draws of the agents in that bin. Therefore, we can talk about the distribution of $\hat{\xi}_t^{s,a}$, that we denote by $\eta_t^{s,a} \in \mathcal{P}(\mathcal{P}(W))$, and collect these across all state-action pairs into the joint object $\eta_t \in \mathcal{P}(\mathcal{P}(W)^{S \times A})$. Finally, to describe the full population transition in this stochastic setting, it is not enough to track where agents are and

Table 1: Notation for the appendix.

Object	Definition	Intuition
$\hat{\xi}_t^{s,a}$	Random element of $\mathcal{P}(W)$	Distribution of noise experienced by agents in (s, a) at time t
$\eta_t^{s,a}$	Distribution of $\hat{\xi}_t^{s,a}$	Describes the probability distribution associated with the noise realizations
η_t	Probability measure in $\mathcal{P}(\mathcal{P}(W)^{S \times A})$	Collects all $\eta_t^{s,a}$ across bins, encoding population-level randomness
ρ_t	$\lambda_t \otimes \hat{\xi}_t^{s,a}$	Joint state-action-noise distribution

what actions they take, one also needs to account for the noise they experience. This motivates the introduction of the joint state-action-noise distribution, defined as the gluing

$$\rho_t = \lambda_t(\mu_t) \otimes \hat{\xi}_t^{s,a} \in \mathcal{P}(S \times A \times W).$$

Intuitively, $\rho_t(\{s, a, w\})$ is the number of agents that are at state s , use action a , and experience the noise realization w . Since the noise realizations are random, $\hat{\xi}_t^{s,a}$ is random and, consequently, also ρ_t is random. At this point, the next population is a pushforward of μ under the system dynamics f_t :

$$\mu_{t+1} = f_{t\#} \rho_t.$$

Different noise models below correspond to different distributional assumptions on $\hat{\xi}_t^{s,a}$, and hence on ρ_t . The objects introduced above are summarized in Table 1.

We impose the following consistency condition on the realized noise model.

Assumption A.1. Fix a timestep t . There exists a Borel set $E_t \subset S \times A$ with $\lambda_t(E_t) = 1$ such that for all $(s, a) \in E_t$ and for every bounded measurable $\varphi : W \rightarrow \mathbb{R}$

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \int_W \varphi(w) d\xi_t(w | s, a). \quad (7)$$

Condition (7) should be read as an unbiased or barycenter condition between the population-level realized noise model $\hat{\xi}_t^{s,a}$ and the true underlying kernel $\xi_t(\cdot | s, a)$. More precisely, the assumption requires that these two descriptions agree after averaging over the realization of $\hat{\xi}_t^{s,a}$. As we will show below, Assumption A.1 is satisfied by several standard realized-kernel constructions. Specifically, In Proposition A.8 we show that it holds true for the noise models presented in Examples A.3- A.6. More generally, Assumption A.1 is satisfied whenever the realized kernel is chosen from a family of probability kernels whose mixture is the nominal kernel $\xi_t(\cdot | s, a)$.

What (7) excludes are biased realized-kernel constructions, as we show in the next example.

Example A.2. Let $W = \{0, 1\}$, $\xi_t = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$. Suppose we artificially bias all noise realization towards 1, i.e., we have

$$\hat{\xi}_t^{s,a} = \delta_1$$

almost surely, then

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W w d\hat{\xi}_t^{s,a}(w) \right] = 1 \neq \frac{1}{2} = \int_W w d\xi_t(w | s, a).$$

Hence, (7) fails.

More generally, it fails whenever the realized population-level kernel systematically overweights some noise realizations relative to the nominal kernel used in the single-agent Bellman equation. In the remainder of this section, we specify this general framework to the individual, local, and global noise models considered in this work (Section A.2). Finally, in Section A.3, we prove a key property of all these noise models: we establish, in Lemma A.9, that the expected cost under the realized empirical kernel $\hat{\xi}_t^{s,a}$ equals the cost under the underlying kernel $\xi_t(\cdot | s, a)$. Ultimately, this unbiasedness property, verified for all four noise models in Proposition A.8, is what allows us to eliminate randomness from the backward recursion in Appendix C and recover the single-agent Bellman equation.

489 A.2 Noise models

490 To start, we consider the case where the noise is i.i.d. over all agents with the same state-action pair.
 491 We start with the case of finitely many agents:

492 **Example A.3** (Individual noise, finitely many agents). *Consider N finite agents, each with its “own”*
 493 *noise sample*

$$w_t^i \sim \xi_t(\cdot | s_t^i, a_t^i).$$

494 *We assume noise draws to be independent across agents and across state-action pairs. Given the state-*
 495 *action distribution λ_t , its support $\text{supp } \lambda_t$ is finite (at most N atoms, since $\lambda_t = \frac{1}{N} \sum_{i=1}^N \delta_{(s_t^i, a_t^i)}$).*
 496 *For each bin $(s, a) \in \text{supp } \lambda_t$, let*

$$N_{s,a} := N \cdot \lambda_t(\{(s, a)\}) \geq 1$$

497 *be the number of agents in that bin. This is a strictly positive integer, determined by λ_t . Define the*
 498 *empirical measure map*

$$\begin{aligned} \Psi_{N_{s,a}}^{s,a} : W^{N_{s,a}} &\rightarrow \mathcal{P}(W) \\ w^{s,a} &\mapsto \frac{1}{N_{s,a}} \sum_{i=1}^{N_{s,a}} \delta_{w_t^i}, \end{aligned}$$

499 *where $w^{s,a} = (w_t^i)_{i \in [N_{s,a}]}$. The realized noise distribution for the state-action pair (s, a) is*

$$\hat{\xi}_t^{s,a} = \Psi_{N_{s,a}}^{s,a}(w^{s,a}) \in \mathcal{P}(W),$$

500 *with distribution*

$$\eta_t^{s,a} = \Psi_{N_{s,a} \#}^{s,a} \left(\bigotimes_{i \in N_{s,a}} \xi_t(\cdot | s, a) \right) \in \mathcal{P}(\mathcal{P}(W)).$$

501 *Since $\text{supp } \lambda_t$ is finite and noise realizations are independent across state-action pairs, the joint law*
 502 *is the finite product*

$$\eta_t = \bigotimes_{(s,a) \in \text{supp } \lambda_t} \eta_t^{s,a} \in \mathcal{P}(\mathcal{P}(W)^{\text{supp } \lambda_t}).$$

503 *We extend η_t to all state-action pairs by setting, for $(s, a) \notin \text{supp } \lambda_t$, $\hat{\xi}_t^{s,a} = \delta_{w^0}$, for an arbitrarily*
 504 *chosen $w^0 \in W$. Clearly, the specific choice of w^0 is inconsequential: since λ_t assigns zero mass to*
 505 *these bins, they do not contribute λ_t -weighted integrals—exactly the type of integrals of our problem*
 506 *setting. The map $(s, a) \mapsto \eta_t^{s,a}$ is Borel measurable: it takes finitely many distinct values (one per*
 507 *element of $\text{supp } \lambda_t$, plus the constant extension outside), and the preimage of each value is a finite*
 508 *union of singletons or the complement of $\text{supp } \lambda_t$, all of which are Borel sets.*

509 **Example A.4** (Individual noise, infinitely many agents). *In the limit case of infinitely many agents,*
 510 *the two-level randomization—each agent drawing a private noise sample and the collection of noise*
 511 *realizations across the population—collapse into a single draw from the deterministic coupling*

$$\rho_t = \lambda_t \otimes \xi_t(\cdot | s, a).$$

512 *Intuitively, the population is “so vast” that its empirical profile of noise realizations matches the*
 513 *underlying noise kernel itself. Rigorously, in the case of finite state and action spaces, we can assume*
 514 *that the number of agents in each state-action pair (s, a) grows to infinity and Varadarajan’s theorem*
 515 *[38, Theorem 11.4.1.] in the space $\mathcal{P}(W)$ guarantees*

$$\hat{\xi}_t^{s,a} \xrightarrow{a.s.} \xi_t, \quad \text{as } N_{s,a} \rightarrow \infty.$$

516 *Consequently, the laws of the realized empirical kernels $\eta_t^{s,a}$ converges weakly to $\delta_{\xi_t(\cdot | s, a)}$ [38,*
 517 *Theorem 9.2.1, Proposition 9.3.5]. We therefore set $\hat{\xi}_t^{s,a} = \xi_t(\cdot | s, a)$ almost surely and*

$$\eta_t^{s,a} = \delta_{\xi_t(\cdot | s, a)} \in \mathcal{P}(\mathcal{P}(W)).$$

518 *For general (possibly uncountable) Polish spaces S and A , we construct the joint law via Kol-*
 519 *mogorov’s extension theorem [39, Theorems 36.1 and 36.2]. Specifically, for each finite subset*
 520 *$J \subset S \times A$, define a probability measure*

$$\eta_t^J = \bigotimes_{(s,a) \in J} \delta_{\xi_t(\cdot | s, a)} \quad \text{on } \mathcal{P}(\mathcal{P}(W)^J).$$

521 These finite-dimensional marginals are consistent (the marginal of η_t^J onto any $J' \subset J$ equals
 522 $\eta_t^{J'}$, since product of Dirac masses marginalize correctly). Thus, Kolmogorov's extension theorem
 523 guarantees that there exists a unique probability measure

$$\eta_t = \bigotimes_{(s,a) \in S \times A} \delta_{\xi_t(\cdot | s, a)},$$

524 and the realized joint state-action-noise distribution collapses to the deterministic coupling $\rho_t =$
 525 $\lambda_t \otimes \xi_t(\cdot | s, a)$.

526 **Example A.5** (Local noise). Suppose the noise realizations are shared among agents with the same
 527 state-action pair. For each fixed $(s, a) \in S \times A$ we draw

$$w_t^{s,a} \sim \xi_t(\cdot | s, a),$$

528 independently across state-action pairs. Then, all agents in bin (s, a) experience the same noise
 529 realization $w_t^i = w_t^{s,a}$. In this framework, we define the empirical measure map

$$\begin{aligned} \Psi^{s,a}: W &\rightarrow \mathcal{P}(W) \\ w_t^{s,a} &\mapsto \delta_{w_t^{s,a}}. \end{aligned}$$

530 The realized noise distribution for bin (s, a) collapses to

$$\hat{\xi}_t^{s,a} = \delta_{w_t^{s,a}},$$

531 with distribution

$$\eta_t^{s,a} = \Psi^{s,a} \# \xi_t(\cdot | s, a) \in \mathcal{P}(\mathcal{P}(W)).$$

532 Since noise samples are independent across state-action pairs, the joint law is constructed by the
 533 same Kolmogorov extension argument as in Example A.4 (applied with $\eta_t^{s,a}$ in place of $\delta_{\xi_t(\cdot | s, a)}$). For
 534 every finite $J \subset S \times A$, the marginal $\eta_t^J = \bigotimes_{(s,a) \in J} \eta_t^{s,a}$ is consistent, yielding the unique extension

$$\eta_t = \bigotimes_{(s,a) \in S \times A} \eta_t^{s,a} \in \mathcal{P}(\mathcal{P}(W)^{S \times A}).$$

535 **Example A.6** (Global noise). Consider the case in which all agents are subject to the same noise
 536 draw $w_t \sim \xi_t \in \mathcal{P}(W)$, regardless of their state and action. Since every state-action pair (s, a)
 537 experiences the same noise realization, we have $\hat{\xi}_t^{s,a} = \delta_{w_t}$ for all (s, a) . We define then the diagonal
 538 embedding

$$\begin{aligned} \Phi: W &\rightarrow \mathcal{P}(W)^{S \times A} \\ w &\mapsto ((s, a) \mapsto \delta_w). \end{aligned}$$

539 The entire random family $\left(\hat{\xi}_t^{s,a} \right)_{(s,a) \in S \times A} = \Phi(w_t)$ is a $\mathcal{P}(W)^{S \times A}$ -valued random variable and its
 540 law is

$$\eta_t = \Phi \# \xi_t \in \mathcal{P}(\mathcal{P}(W)^{S \times A}).$$

541 In words, $\xi_t \in \mathcal{P}(W)$ is pushed forward through Φ to a measure on $\mathcal{P}(W)^{S \times A}$, concentrated on the
 542 diagonal $\{\Phi(w) : w \in W\}$. We refer the reader to Remark A.7 for a discussion on the well-posedness
 543 of Φ .

544 **Remark A.7** (Well-posedness of Example A.6). We verify here that the map $\Phi: W \rightarrow \mathcal{P}(W)^{S \times A}$ is
 545 well-defined and a Borel measurable map between Polish spaces, so that the pushforward $\eta_t = \Phi \# \xi_t$
 546 is well-defined.

547 **Step 1:** Each coordinate map is continuous. We start by noting that each coordinate map $w \mapsto$
 548 $\delta_w: W \rightarrow \mathcal{P}(W)$ is continuous with respect to the weak topology on $\mathcal{P}(W)$ [39, Example 2.1].

549 **Step 2:** Φ is Borel measurable. To show that the full map Φ is Borel measurable we consider two
 550 cases depending on whether $S \times A$ is countable or uncountable. In the countable case, $\mathcal{P}(W)^{S \times A}$
 551 is a countable product of Polish spaces, which is again a Polish space [40, Proposition 3.3] under
 552 the product topology. A map into a countable product is measurable if and only if each coordinate
 553 projection composed with it is measurable, which holds by Step 1. So Φ is Borel measurable. If,

instead, $S \times A$ is uncountable, then $\mathcal{P}(W)^{S \times A}$ might fail to be Polish. However, this does not pose issues as the image of Φ is contained in the diagonal subset

$$\mathcal{D} = \{\Phi(w) : w \in W\} = \{(s, a) \mapsto \delta_w : w \in W\} \subset \mathcal{P}(W)^{S \times A}.$$

The map $\Phi : W \rightarrow \mathcal{D}$ is a homeomorphism: both Φ and its inverse $\Phi^{-1} : \mathcal{D} \rightarrow W$ (which evaluates any coordinate and then inverts $w \mapsto \delta_w$) are continuous. Hence, $\mathcal{D}$ is homeomorphic to W , which is Polish. In particular, $\mathcal{D}$ is a Polish space in the subspace topology inherited from $\mathcal{P}(W)^{S \times A}$. Since $\Phi : W \rightarrow \mathcal{D}$ is continuous between Polish spaces it is Borel measurable, and $\eta_t = \Phi_{\#} \xi_t$ is a well-defined Borel probability measure on $\mathcal{D}$ and so also on $\mathcal{P}(W)^{S \times A}$. We finally observe that η_t is fully concentrated on $\mathcal{D}$, so all subsequent integrals only involve elements of $\mathcal{D}$ and the uncountability of $S \times A$ does not pose an issue.

We now show that these canonical noise models satisfy Assumption A.1.

Proposition A.8 (The noise models satisfy Assumption A.1). *For each of the noise models in Examples A.3-A.6, the realized kernel $\hat{\xi}_t^{s,a}$ and its distribution η_t satisfy Assumption A.1. That is, there exists a Borel set $E_t \in S \times A$ with $\lambda_t(E_t) = 1$ such that for all $(s, a) \in E_t$ and for every bounded measurable $\varphi : W \rightarrow \mathbb{R}$ simultaneously,*

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \int_W \varphi(w) d\xi_t(w | s, a).$$

Proof. We prove the statement for each noise model separately.

Individual noise (Example A.3). Fix $(s, a) \in \text{supp } \lambda_t$, hence $N_{s,a} \geq 1$. By construction,

$$\hat{\xi}_t^{s,a} = \frac{1}{N_{s,a}} \sum_{i=1}^{N_{s,a}} \delta_{w_t^i}$$

where w_t^i are i.i.d with law $\xi_t(\cdot | s, a)$. Hence, for any bounded measurable $\varphi : W \rightarrow \mathbb{R}$,

$$\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) = \frac{1}{N_{s,a}} \sum_{i=1}^{N_{s,a}} \varphi(w_t^i)$$

and therefore, by linearity of expectation,

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \frac{1}{N_{s,a}} \sum_{i=1}^{N_{s,a}} \mathbb{E}_{\hat{\xi}_t^{s,a}} [\varphi(w_t^i)] = \int_W \varphi(w) d\xi_t(w | s, a).$$

thus (7) holds on $E_t = \text{supp } \lambda_t$, which clearly satisfies $\lambda_t(E_t) = 1$.

Infinite-population model (Example A.4). By construction $\hat{\xi}_t^{s,a} = \xi_t(\cdot | s, a)$ almost surely for every $(s, a) \in S \times A$. Therefore, for every bounded measurable φ

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \int_W \varphi(w) d\xi_t(w | s, a).$$

Thus, (7) holds with $E_t = S \times A$.

Local noise (Example A.5). For each fixed (s, a) , one draw $w_t^{s,a} \sim \xi_t(\cdot | s, a)$ independently across state-action pairs, and the realized noise distribution is $\hat{\xi}_t^{s,a} = \delta_{w_t^{s,a}}$. So, for every bounded measurable φ

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \varphi(w) d\hat{\xi}_t^{s,a}(w) \right] = \mathbb{E}_{\hat{\xi}_t^{s,a}} [\varphi(w_t^{s,a})] = \int_W \varphi(w) d\xi_t(w | s, a).$$

Thus, (7) holds again with $E_t = S \times A$.

Global-noise (Example A.6). This case is identical to the local noise one, with the same draw w_t shared across all (s, a) . $\square$

582 A.3 Averaging property of realized noise kernels

583 We now show that even if $\hat{\xi}_t^{s,a}$ is random, the expected cost computed under its law η_t equals the
 584 cost computed directly under the underlying kernel $\xi_t(\cdot | s, a)$. In other words, replacing the random
 585 empirical noise law $\hat{\xi}_t^{s,a}$ by the underlying kernel $\xi_t(\cdot | s, a)$ is unbiased in expectation. Among
 586 others, this lemma will allow us to remove randomness from the backward recursion in the proof of
 587 Theorem 3.1, Part I.

588 **Lemma A.9** (Averaging realized kernels). *Let S, A, W be Polish spaces and fix a timestep t . Let*
 589 *λ_t denote the joint state-action distribution and $\xi_t : S \times A \rightarrow \mathcal{P}(W)$ be a stochastic kernel. For*
 590 *each (s, a) , let $\hat{\xi}_t^{s,a}$ be a $\mathcal{P}(W)$ -valued random variable with distribution $\eta_t^{s,a} \in \mathcal{P}(\mathcal{P}(W))$, so that*
 591 *Assumption A.1 holds true and $(s, a) \mapsto \eta_t^{s,a}$ is measurable. Then, for every nonnegative measurable*
 592 *$c : S \times A \times W \rightarrow \mathbb{R}_{\geq 0}$,*

$$\int_{S \times A} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c(s, a, w) d\hat{\xi}_t^{s,a}(w) \right] d\lambda_t(s, a) = \int_{S \times A} \int_W c(s, a, w) d\xi_t(w | s, a) d\lambda_t(s, a). \quad (8)$$

593 *Proof.* For $n \in \mathbb{N}$, define

$$c_n(s, a, w) := \min\{c(s, a, w), n\}.$$

594 By construction, each c_n is bounded and measurable, and $c_n \uparrow c$ pointwise.

595 To start, we prove that the two integrands in (8) are measurable. For the left-hand side, for each n , the
 596 map $(s, a, \zeta) \mapsto \int_W c_n d\zeta$ is Borel measurable on $S \times A \times \mathcal{P}(W)$ and continuous in ζ . This follows
 597 from a standard monotone-class argument, starting from indicators of measurable rectangles and
 598 using the fact that $\zeta \mapsto \zeta(B)$ is Borel measurable for every Borel set $B \in W$ [29, Proposition 7.28].
 599 Since $(s, a) \mapsto \eta_t^{s,a}$ is measurable, standard measurability arguments of the stochastic kernel yield
 600 measurability

$$(s, a) \mapsto \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c_n(s, a, w) d\hat{\xi}_t^{s,a}(w) \right].$$

601 Letting $n \uparrow \infty$, the monotone convergence theorem [39, Theorem 16.2] yields the measurability of

$$(s, a) \mapsto \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c(s, a, w) d\hat{\xi}_t^{s,a}(w) \right].$$

602 For the right-hand side, instead, the measurability of

$$(s, a) \mapsto \int_W c(s, a, w) d\xi_t(w | s, a).$$

603 follows from standard properties of stochastic kernels [29, Propositions 7.28 and 7.29].

604 We now prove the equality (8). Fix n , for every $(s, a) \in E_t$, apply (7) with the bounded measurable
 605 test function $\varphi_{s,a}(w) := c_n(s, a, w)$, viewed as a function of w alone for fixed (s, a) . This gives

$$\mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c_n(s, a, w) d\hat{\xi}_t^{s,a}(w) \right] = \int_W c_n(s, a, w) d\xi_t(w | s, a)$$

606 for all $(s, a) \in E_t$. Since $\lambda_t(E_t) = 1$, integrating both sides with respect to λ_t gives

$$\int_{S \times A} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W c_n(s, a, w) d\hat{\xi}_t^{s,a}(w) \right] d\lambda_t(s, a) = \int_{S \times A} \int_W c_n(s, a, w) d\xi_t(w | s, a) d\lambda_t(s, a).$$

607 Finally, let $n \rightarrow \infty$. Since $c_n \uparrow c$, the monotone convergence theorem yields (8). $\square$

608 B Preliminary lemmata

609 We now collect the auxiliary results required for the main proof in Appendix C.

610 **Measurability convention.** Throughout the appendix, primitive data such as the dynamics, stage
 611 costs, and terminal cost are assumed to be Borel measurable, as in the main text. Since every
 612 nonnegative Borel measurable function is lower semianalytic, these data also satisfy the analytic-
 613 measurability assumptions used in the auxiliary results below. We use lower semianalytic functions

because the dynamic programming recursion involves pointwise infima over the action space, and such infima need not be Borel measurable on general Polish spaces; see [29] for a detailed discussion. Accordingly, value functions and pointwise Bellman infima are interpreted as lower semianalytic, hence universally measurable, functions. This is sufficient for the integrals and measurable-selection arguments used in the proof of Theorem 3.1. We work at this level of generality to simultaneously capture systems with finite and continuous state and actions spaces. Nevertheless, when all spaces are finite, measurability is automatic and all measurability arguments can therefore be skipped.

We begin by recalling from [25] a result that lets us “move” pushforward from the set of coupling into the transportation cost, and vice versa.

Lemma B.1 (Pushforward and optimal transport [25, Lemma 6.1]). *Given a transportation cost $j_{t+1} : S \times Z \rightarrow \mathbb{R}_{\geq 0}$, a map $f_t : S \times A \times W \rightarrow S$, and probability measure $\rho_t \in \mathcal{P}(S \times A \times W)$, it holds that*

$$\inf_{(\text{proj}_{S \times A \times W}^{S \times A \times W \times Z})_{\#} \gamma = \rho_t} \int_{S \times A \times W \times Z} j_{t+1} \circ (f_t \times \text{Id}_Z) d\gamma = \inf_{(\text{proj}_S^{S \times Z})_{\#} \tilde{\gamma} = f_{t\#} \rho_t} \int_{S \times Z} j_{t+1} d\tilde{\gamma},$$

where Id_Z the identity map over Z .

In our setting, this lemma tells us that evaluating the cost after applying the system dynamics is equivalent to pushing forward the distribution first. Thus, this means we can “apply the dynamics first” and then evaluate the population-level cost without changing the result.

Next, we recall how to merge additive costs into a single transport objective.

Lemma B.2 (Sum of optimal transport discrepancies [25, Lemma 6.2]). *Given transportation costs $c_1 : X \rightarrow \mathbb{R}_{\geq 0}$, $c_2 : X \times Z \rightarrow \mathbb{R}_{\geq 0}$, and probability measures $\rho \in \mathcal{P}(X)$, $\nu \in \mathcal{P}(Z)$, it holds that*

$$\mathbb{E}_{\rho} [c_1] + \mathcal{K}[c_2](\rho, \nu) = \mathcal{K}[c](\rho, \nu),$$

with $c : X \times Z \rightarrow \mathbb{R}_{\geq 0}$ defined as $c(x, z) = c_1(x) + c_2(x, z)$.

This result allows us to collapse stage costs and terminal transport costs into a unified expression, which is crucial for dynamic programming formulations.

We now introduce stochasticity which, in our setting, will appear due to the stochastic dynamics. Through a standard argument, we show that we can safely exchange expectation over random noise with the infimum over couplings, at the price of an inequality.

Lemma B.3 (Infimum and expectation). *Let $\lambda \in \mathcal{P}(X)$ and $\nu \in \mathcal{P}(Z)$. Consider $\hat{\xi} = (\hat{\xi}^x)_{\{x \in X\}} \in \mathcal{P}(W)^X$ be a random family with $\hat{\xi}^x \in \mathcal{P}(W)$, and denote with $\eta \in \mathcal{P}(\mathcal{P}(W)^X)$ its distribution. Let $c : X \times W \times Z \rightarrow \mathbb{R}_{\geq 0}$ be measurable. Then*

$$\begin{aligned} \mathbb{E}_{\hat{\xi}} \left[\inf_{\gamma \in \Gamma(\lambda, \nu)} \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma(x, z) \right] \\ \leq \inf_{\gamma \in \Gamma(\lambda, \nu)} \mathbb{E}_{\hat{\xi}} \left[\int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma(x, z) \right]. \end{aligned}$$

Proof. Fix any $\gamma \in \Gamma(\lambda, \nu)$. Pointwise in $\hat{\xi}$,

$$\inf_{\gamma \in \Gamma(\lambda, \nu)} \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma(x, z) \leq \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma(x, z).$$

Taking expectations over $\hat{\xi}$ preserves the inequality. Since the right-hand side holds for every coupling γ , taking the infimum over $\Gamma(\lambda, \nu)$ yields the claim. $\square$

With this inequality, randomness can be “pushed inside” the transport problem, a step that will later allow us to handle noisy transitions.

The next tool, recalled from [25], refines our handling of couplings: it shows how an optimizer can be disintegrated with respect to one marginal, yielding conditional distributions.

649 **Lemma B.4** (Disintegration of the optimizer [25, Lemma 6.4]). *Given a transportation cost $c : S \times A \times Z \rightarrow \mathbb{R}_{\geq 0}$ and probability measures $\mu \in \mathcal{P}(S)$, $\nu \in \mathcal{P}(Z)$, it holds that*

$$\inf_{(\text{proj}_{S \times Z}^{S \times A \times Z})_{\#} \gamma \in \Gamma(\mu, \nu)} \int_{S \times A \times Z} c(s, a, z) d\gamma(s, a, z) = \inf_{\gamma' \in \Gamma(\mu, \nu)} \inf_{\{\alpha^{s, z}\} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s, z}(a) d\gamma'(s, z), \quad (9)$$

651 where $\Lambda(A) := \{\{\alpha^{s, z}\}_{(s, z) \in S \times Z} \subseteq \mathcal{P}(A) \mid S \times Z \ni (s, z) \mapsto \alpha^{s, z} \in \mathcal{P}(A) \text{ Borel}\}$.

652 This separation will allow us to express the population-level policy as a distribution over conditional agent-level actions.

654 Next, we study more in detail the right-hand side in (9), showing that the infimum over the family $\{\alpha^{s, z}\} \in \Lambda(A)$ can be “pushed” into the integral.

656 **Lemma B.5** (Reduction to pointwise infimum over deterministic policies). *Let S, A, Z be Polish spaces and let $c : S \times A \times Z \rightarrow \mathbb{R}_{\geq 0}$. Let $\mu \in \mathcal{P}(S)$, $\nu \in \mathcal{P}(Z)$ and $\gamma \in \Gamma(\mu, \nu)$. Define*

$$\Lambda(A) := \{\{\alpha^{s, z}\}_{(s, z) \in S \times Z} \subseteq \mathcal{P}(A) \mid S \times Z \ni (s, z) \mapsto \alpha^{s, z} \in \mathcal{P}(A) \text{ is Borel measurable}\}.$$

658 Then, the function $(s, z) \mapsto \inf_{a \in A} c(s, a, z)$ is universally measurable, and for every coupling $\gamma \in \Gamma(\mu, \nu)$,

$$\inf_{\alpha^{s, z} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s, z}(a) d\gamma(s, z) = \int_{S \times Z} \inf_{a \in A} c(s, a, z) d\gamma(s, z). \quad (10)$$

660 *Proof.* We start by assessing the measurability of the pointwise infimum. Since c is lower semianalytic (see the measurability convention at the beginning of Appendix B) and A is a Polish space, the function $(s, z) \mapsto \inf_{a \in A} c(s, a, z)$ is lower semianalytic by [29, Proposition 7.47], hence universally measurable. In particular, the right-hand side of (10) is a well-defined element of $[0, +\infty]$.

664 We then prove (10) by showing “ $\leq$ ” and “ $\geq$ ” separately.

665 **Lower bound “ $\geq$ ”.** For any $\{\alpha^{s, z}\} \in \Lambda(A)$ and every $(s, z) \in S \times Z$, since $\alpha^{s, z} \in \mathcal{P}(A)$ is a probability measure

$$\int_A c(s, a, z) d\alpha^{s, z}(a) \geq \inf_{a \in A} c(s, a, z).$$

667 Integrating both sides with respect to γ and taking the infimum over all $\{\alpha^{s, z}\} \in \Lambda(A)$ leads to

$$\inf_{\{\alpha^{s, z}\} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s, z}(a) d\gamma(s, z) \geq \int_{S \times Z} \inf_{a \in A} c(s, a, z) d\gamma(s, z).$$

668 **Upper bound “ $\leq$ ”.** Let $\varepsilon > 0$. Since $c \geq 0$ is lower semianalytic and A is a Polish space, [29, Proposition 7.50] guarantees the existence of a universally measurable ε -optimal selector $a^\varepsilon : S \times Z \rightarrow A$ satisfying

$$c(s, a^\varepsilon(s, z), z) \leq \inf_{a \in A} c(s, a, z) + \varepsilon \quad \forall (s, z) \in S \times Z.$$

671 Now, since γ is a σ -finite measure on $S \times Z$, by [29, Lemma 7.27] there exists a Borel measurable map $b^\varepsilon : S \times Z \rightarrow A$ with $b^\varepsilon = a^\varepsilon$ γ -almost everywhere, so that

$$c(s, b^\varepsilon(s, z), z) \leq \inf_{a \in A} c(s, a, z) + \varepsilon \quad \gamma\text{-almost everywhere.}$$

673 The Dirac embedding $A \rightarrow \mathcal{P}(A)$ defined by $a \mapsto \delta_a$ is continuous in the weak topology [39, Example 2.1], hence Borel measurable. Therefore, the composition $(s, z) \mapsto \delta_{b^\varepsilon(s, z)}$ is Borel measurable, and so is the family $\{\delta_{b^\varepsilon(s, z)}\}_{(s, z) \in S \times Z} \in \Lambda(A)$. Thus,

$$\begin{aligned} \inf_{\{\alpha^{s, z}\} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s, z}(a) d\gamma(s, z) &\leq \int_{S \times Z} c(s, b^\varepsilon(s, z), z) d\gamma(s, z) \\ &\leq \int_{S \times Z} \inf_{a \in A} c(s, a, z) d\gamma(s, z) + \varepsilon. \end{aligned}$$

676 Since $\varepsilon > 0$ is arbitrary, we conclude

$$\inf_{\{\alpha^{s,z}\} \in \Lambda(A)} \int_{S \times Z} \int_A c(s, a, z) d\alpha^{s,z}(a) d\gamma(s, z) \leq \int_{S \times Z} \inf_{a \in A} c(s, a, z) d\gamma(s, z).$$

677 This completes the proof of the upper bound and, with it, the proof of the lemma. $\square$

678 This lemma will allow us to relate the dynamic programming over probability measures with the
679 standard dynamic programming over the state and action spaces.

680 Finally, we present a lemma that upper bounds the transport cost when replacing “full couplings”
681 with “marginal plans”. This inequality will be used to simplify the optimization by relaxing the
682 coupling constraints, enabling us to separate noise from the population-level allocation.

683 **Lemma B.6** (Suboptimality of marginal plans). *Given a transportation cost $c : X \times W \times Z \rightarrow \bar{\mathbb{R}}_{\geq 0}$
684 and probability measures $\rho \in \mathcal{P}(X \times W)$, $\nu \in \mathcal{P}(Z)$, it holds that*

$$\inf_{\gamma \in \Gamma(\rho, \nu)} \int_{X \times W \times Z} c(x, w, z) d\gamma(x, w, z) \leq \inf_{\gamma_1 \in \Gamma(\lambda, \nu)} \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma_1(x, z), \quad (11)$$

685 where $\rho = \lambda \otimes \hat{\xi}^x$ is a disintegration of ρ with respect to its first marginal λ .

686 *Proof.* Consider the disintegration of ρ with respect to its first marginal $\lambda = (\text{proj}_{X \times W}^{X \times W})_{\#} \rho \in \mathcal{P}(X)$,
687 $\rho = \lambda \otimes \hat{\xi}^x$. Let $\gamma_1 \in \Gamma(\lambda, \nu)$ and consider the gluing $\bar{\gamma} = \gamma_1 \otimes \hat{\xi}^x \in \mathcal{P}(X \times Z \times W)$. To “reorder”
688 marginals, consider the permutation

$$\sigma : X \times Z \times W \rightarrow X \times W \times Z, \quad \sigma(x, z, w) = (x, w, z),$$

689 and set $\gamma = \sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \in \mathcal{P}(X \times W \times Z)$.

690 We now show that γ is a coupling between ρ and ν and, thus, a valid upper bound for the infimum
691 in (11). We check each marginal by testing against an arbitrary bounded measurable function. Let
692 $\varphi : X \times W \rightarrow \mathbb{R}$ be bounded and measurable, we have

$$\begin{aligned} \int_{X \times W} \varphi(x, w) d \left(\left(\text{proj}_{X \times W}^{X \times W \times Z} \right)_{\#} \gamma \right) (x, w) \\ &= \int_{X \times W \times Z} \varphi \left(\text{proj}_{X \times W}^{X \times W \times Z}(x, w, z) \right) d\gamma(x, w, z) \\ &= \int_{X \times W \times Z} \varphi(\text{proj}_{X \times W}^{X \times W \times Z}(x, w, z)) d \left(\sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \right) (x, w, z) \\ &= \int_{X \times Z \times W} \varphi \left(\text{proj}_{X \times W}^{X \times W \times Z}(\sigma(x, z, w)) \right) d(\gamma_1 \otimes \hat{\xi}^x)(x, z, w) \\ &= \int_{X \times Z} \int_W \varphi(x, w) d\hat{\xi}^x(w) d\gamma_1(x, z) \\ &= \int_X \int_W \varphi(x, w) d\hat{\xi}^x(w) d\lambda(x). \end{aligned}$$

693 For every bounded and measurable φ , since $\hat{\xi}^x$ is the disintegration of ρ on W given x , and $\gamma_1 \in$
694 $\Gamma(\lambda, \nu)$ has first marginal λ , Fubini–Tonelli [39, Theorem 13.8] gives

$$\int_X \int_W \varphi(x, w) d\hat{\xi}^x(w) d\lambda(x) = \int_{X \times W} \varphi(x, w) d(\lambda \otimes \hat{\xi}^x)(x, w) = \int_{X \times W} \varphi(x, w) d\rho(x, w).$$

695 Hence $(\text{proj}_{X \times W}^{X \times W \times Z})_{\#} \gamma = \rho$. Similarly, for the marginal over Z , let $\psi: Z \rightarrow \mathbb{R}$ be bounded and
 696 measurable. Then,

$$\begin{aligned} & \int_Z \psi(z) d \left(\left(\text{proj}_Z^{X \times W \times Z} \right)_{\#} \gamma \right) (z) \\ &= \int_{X \times W \times Z} \psi \left(\text{proj}_Z^{X \times W \times Z} (x, w, z) \right) d\gamma(x, w, z) \\ &= \int_{X \times W \times Z} \psi \left(\text{proj}_Z^{X \times W \times Z} (x, w, z) \right) d \left(\sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \right) (x, w, z) \\ &= \int_{X \times Z \times W} \psi \left(\text{proj}_Z^{X \times W \times Z} (\sigma(x, z, w)) \right) d \left(\gamma_1 \otimes \hat{\xi}^x \right) (x, z, w) \\ &= \int_{X \times Z} \int_W \psi(z) d\hat{\xi}^x(w) d\gamma_1(x, z). \end{aligned}$$

697 Since $\int_W d\hat{\xi}^x(w) = 1$ for all x and ν is the marginal over Z of $\gamma_1 \in \Gamma(\lambda, \nu)$, this reduces to

$$\int_{X \times Z} \psi(z) d\gamma_1(x, z) = \int_Z \psi(z) d \left(\left(\text{proj}_Z^{X \times W \times Z} \right)_{\#} \gamma \right) (z) = \int_Z \psi(z) d\nu(z),$$

698 Hence $(\text{proj}_Z^{X \times W \times Z})_{\#} \gamma = \nu$. Overall, we have proven that, for all $\gamma_1 \in \Gamma(\lambda, \nu)$, $\gamma =$
 699 $\sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \in \Gamma(\rho, \nu)$.

700 We are now ready to prove (11). Fix $\gamma_1 \in \Gamma(\lambda, \nu)$. Since the coupling $\gamma = \sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \in \Gamma(\rho, \nu)$
 701 is a valid candidate for the left-hand side. Thus, we obtain the upper bound

$$\begin{aligned} \inf_{\tilde{\gamma} \in \Gamma(\rho, \nu)} \int_{X \times W \times Z} c(x, w, z) d\tilde{\gamma}(x, w, z) &\leq \int_{X \times W \times Z} c(x, w, z) d\gamma(x, w, z) \\ &= \int_{X \times W \times Z} c(x, w, z) d \left(\sigma_{\#}(\gamma_1 \otimes \hat{\xi}^x) \right) (x, w, z) \\ &= \int_{X \times Z \times W} c(x, w, z) d \left(\gamma_1 \otimes \hat{\xi}^x \right) (x, z, w) \\ &= \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma_1(x, z), \end{aligned}$$

702 where the last equality uses disintegration and Tonelli's theorem to get. Since $\gamma_1 \in \Gamma(\lambda, \nu)$ is
 703 arbitrary, we may take infima to obtain

$$\inf_{\tilde{\gamma} \in \Gamma(\rho, \nu)} \int c(x, w, z) d\tilde{\gamma}(x, w, z) \leq \inf_{\gamma_1 \in \Gamma(\lambda, \nu)} \int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma_1(x, z),$$

704 which is the desired statement. □

705 Through the following example, we show that the inequality (11) is generally strict.

706 **Example B.7** (Strictness of Lemma B.6). *Let $X = W = Z = \mathbb{R}$. Set $\lambda = \mathcal{N}(0, 1)$ on X . For each*
 707 *$x \in X$, let $\hat{\xi} = \hat{\xi}^x = \mathcal{N}(0, 1)$ on W , independent of x . Then, the product measure $\rho = \lambda \otimes \hat{\xi}$ is the*
 708 *law of a pair (x, w) with $x, w \stackrel{i.i.d.}{\sim} \mathcal{N}(0, 1)$. Let $\nu = \mathcal{N}(0, 1)$ on Z and define the cost*

$$c(x, w, z) = (w - z)^2.$$

709 To evaluate the left-hand side in (11) in Lemma B.6, we aim to solve

$$\inf_{\gamma \in \Gamma(\rho, \nu)} \int_{\mathbb{R}^3} c(x, w, z) d\gamma(x, w, z).$$

710 To do so, we show that there exists a plan $\tilde{\gamma} \in \Gamma(\rho, \nu)$ that attains a cost of 0, which, by non-negativity
 711 of the cost, implies optimality. Consider $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, w) = (x, w, w)$ and set

712 $\tilde{\gamma} = T_{\#}\rho$. By construction, its (x, w) -marginal is ρ and its z -marginal is given by $w \sim \mathcal{N}(0, 1) = \nu$,
 713 hence $\tilde{\gamma} \in \Gamma(\rho, \nu)$. Under this coupling

$$\int_{\mathbb{R}^3} (w - z)^2 d\tilde{\gamma}(x, w, z) = \int_{\mathbb{R}^3} (w - w)^2 d\rho(x, w) = 0$$

714 and so

$$\inf_{\gamma \in \Gamma(\rho, \nu)} \int_{\mathbb{R}^3} c(x, w, z) d\gamma(x, w, z) = 0.$$

715 We now consider the right-hand side in (11). For any $\gamma_1 \in \Gamma(\lambda, \nu)$, we evaluate

$$\int_{X \times Z} \int_W c(x, w, z) d\hat{\xi}^x(w) d\gamma_1(x, z).$$

716 Since the disintegration of ρ is $\hat{\xi} = \mathcal{N}(0, 1)$ (in particular, it is independent of x), we have

$$\int_{\mathbb{R}} c(x, w, z) d\hat{\xi}(w) = \int_{\mathbb{R}} (w - z)^2 d\hat{\xi}(w) = \int_{\mathbb{R}} w^2 d\hat{\xi}(w) - 2z \int_{\mathbb{R}} w d\hat{\xi}(w) + z^2 = 1 + z^2,$$

717 and so

$$\begin{aligned} \inf_{\gamma_1 \in \Gamma(\lambda, \nu)} \int_{\mathbb{R}^2} \int_{\mathbb{R}} c(x, w, z) d\hat{\xi}(w) d\gamma_1(x, z) &= \inf_{\gamma_1 \in \Gamma(\lambda, \nu)} \int_{\mathbb{R}^2} 1 + z^2 d\gamma_1(x, z) \\ &= 1 + \int_{\mathbb{R}} z^2 d\nu(z) \\ &= 2, \end{aligned}$$

718 where the last equality follows from the fact that the second marginal of γ_1 is $\nu = \mathcal{N}(0, 1)$ and thus
 719 its second moment is 1. Clearly, the left- and right hand sides do not agree, and the inequality (11) is
 720 strict in general.

721 Example B.7 shows that the inequality of Lemma B.6—and hence the bound of Theorem 3.1,
 722 Part I—is generally strict. Informally, the gap corresponds to the value of being able to match noise
 723 realizations directly to targets, rather than averaging over them. In control problems where noise is
 724 small relative to the nominal cost (as in the low-noise regime of our grid-world experiments), we
 725 expect this gap to be small.

726 C Proofs of Section 3

727 We now prove the main results presented in Section 3. Recall the single-agent problem and the
 728 dynamic programming for the cost-to-go function in (2):

$$\begin{aligned} j_T(s_T, z) &= g_T(s_T, z); \\ j_t(s_t, z) &= \inf_{a_t \in A} \mathbb{E}_{w_t} [g_t(s_t, a_t, w_t) + j_{t+1}(f_t(s_t, a_t, w_t), z)]. \end{aligned}$$

729 C.1 Proof of Theorem 3.1

730 We aim to show that the population level cost-to-go function in Problem 2 admits as upper bound an
 731 optimal transport problem whose cost is the single-agent cost-to-go function in (2). Concretely, we
 732 show by backward induction that

$$J_t(\mu_t, \nu) \leq \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int j_t(s, z) d\gamma(s, z).$$

733 We prove the theorem under the more general setting where Assumption A.1 is in place. As shown
 734 in Proposition A.8, this assumption is satisfied by all three noise models—individual, local, and
 735 global—considered in the main body of the paper. This will allow us to use Lemma A.9 in the proof
 736 of the theorem.

737 We start with the proof of Part I of the theorem.

738 *Proof.* The proof proceeds by induction, in three steps. First, we use the definition of the cost-to-go
 739 function and the induction hypothesis at time $t + 1$. Second, we express the conditional expectations
 740 over noise realizations in terms of the averaged single-agent cost. Finally, we reorganize the infima
 741 over couplings by means of Lemma B.4, which allows us to recover the desired distributional dynamic
 742 program and conclude the induction.

743 The base case of the induction is straightforward. By definition $J_T(\mu_T) = \mathcal{K}[g_T](\mu_T, \nu)$. Since
 744 $j_T(s, z) = g_T(s, z)$, we have $\mathcal{K}[j_T](\mu_T, \nu) = \mathcal{K}[g_T](\mu_T, \nu) = J_T(\mu_T)$. Fix $t < T$, using the
 745 induction hypothesis at time $t + 1$, i.e.,

$$J_{t+1}(f_{t\#}\rho_t, \nu) \leq \inf_{\gamma \in \Gamma(f_{t\#}\rho_t, \nu)} \int_{S \times Z} j_{t+1}(s^+, z) d\gamma,$$

746 the backward recursion for the cost-to-go function at time t gives

$$\begin{aligned} J_t(\mu_t, \nu) &= \inf_{(\text{proj}_S^{S \times A})_{\#} \lambda_t = \mu_t} \mathbb{E}_{\hat{\xi}_t} [\mathbb{E}_{\rho_t} [g_t(s, a, w)] + J_{t+1}(f_{t\#}\rho_t, \nu)] \\ &\leq \inf_{(\text{proj}_S^{S \times A})_{\#} \lambda_t = \mu_t} \mathbb{E}_{\hat{\xi}_t} \left[\underbrace{\int_{S \times A \times W} g_t(s, a, w) d\rho_t + \inf_{\gamma \in \Gamma(f_{t\#}\rho_t, \nu)} \int_{S \times Z} j_{t+1}(s^+, z) d\gamma}_{(*)} \right]. \quad (12) \end{aligned}$$

747 We now focus on the term denoted by $(*)$. We analyze $(*)$ pointwise for each realization of $\hat{\xi}_t$ before
 748 taking expectations. By Lemma B.1, we can “push” the dynamics into the transportation cost, so that

$$(*) = \int_{S \times A \times W} g_t(s, a, w) d\rho_t + \inf_{\gamma' \in \Gamma(\rho_t, \nu)} \int_{S \times A \times W \times Z} j_{t+1}(f_t(s, a, w), z) d\gamma'.$$

749 Since ρ_t is the first marginal of γ' , Lemma B.2, with $X = S \times A \times W$, $c_1 = g_t$ and $c_2 =$
 750 $j_{t+1} \circ (f_t \times \text{Id}_Z)$, allows to merge the two costs into a single transport objective on $S \times A \times W \times Z$.
 751 We therefore have

$$(*) = \inf_{\gamma' \in \Gamma(\rho_t, \nu)} \int_{S \times A \times W \times Z} g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\gamma'.$$

752 We now use Lemma B.6 with $X = S \times A$, $\rho_t = \lambda_t \otimes \hat{\xi}_t^{s,a}$, and $c(s, a, w, z) = g_t(s, a, w) +$
 753 $j_{t+1}(f_t(s, a, w), z)$, to replace the couplings of the full state-action-noise distribution ρ_t with cou-
 754 plings of its $(S \times A)$ -marginal λ_t . This yields the inequality

$$(*) \leq \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} d\tilde{\gamma}.$$

755 This relaxation removes the dependence on the realized noise from the optimal assignment, thus
 756 restricting it to depend only on (s, a) .

757 We now plug the obtained upper bound for $(*)$ inside the expectation over $\hat{\xi}_t^{s,a}$, as in (12). Since λ_t
 758 is fixed and the infimum is now over couplings $\tilde{\gamma} \in \Gamma(\lambda_t, \nu)$ that do not depend on $\hat{\xi}_t^{s,a}$, Lemma B.3
 759 applies, allowing us to interchange expectation and infimum and yielding the upper bound

$$\begin{aligned} \mathbb{E}_{\hat{\xi}_t} [(*)] &\leq \mathbb{E}_{\hat{\xi}_t} \left[\inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} d\tilde{\gamma} \right] \\ &\leq \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \mathbb{E}_{\hat{\xi}_t} \left[\int_{S \times A \times Z} \int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} d\tilde{\gamma} \right] \\ &= \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} \right] d\tilde{\gamma}, \end{aligned}$$

760 where the last equality follows from Tonelli’s theorem, which applies since the stage cost g_t and the
 761 cost-to-go functions j_{t+1} are nonnegative. Informally, the second inequality is precisely when we
 762 transition from “choose the assignment after seeing the realized noise” to “choose one assignment

763 against average noise". We now focus on the inner expectation. In particular, consider a fixed $\tilde{\gamma} \in$
 764 $\Gamma(\lambda_t, \nu)$. We can disintegrate the coupling with respect to its $(S \times A)$ -marginal λ_t as $\tilde{\gamma} = \lambda_t \otimes \kappa_{s,a}$
 765 where $\kappa_{s,a}$ is a (conditional) probability over Z . By non-negativity of all functions, Tonelli's theorem
 766 gives

$$\begin{aligned} & \int_{S \times A \times Z} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\hat{\xi}_t^{s,a} \right] d\tilde{\gamma} \\ &= \int_{S \times A} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \int_Z g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\kappa_{s,a} d\hat{\xi}_t^{s,a} \right] d\lambda_t. \end{aligned}$$

767 We can apply Lemma A.9 (treating (s, a) as parameters) to the function

$$\tilde{c}(s, a, w) := \int_Z g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\kappa_{s,a}(z).$$

768 Note, in particular, that Lemma A.9 can be applied since $\kappa_{s,a}$ is fixed and not a function of the noise
 769 distribution $\hat{\xi}_t^{s,a}$. Thus, we get

$$\begin{aligned} & \int_{S \times A} \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_W \int_Z g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\kappa_{s,a} d\hat{\xi}_t^{s,a} \right] d\lambda_t \\ &= \int_{S \times A} \mathbb{E}_w \left[\int_Z g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z) d\kappa_{s,a} \right] d\lambda_t \\ &= \int_{S \times A \times Z} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma} \end{aligned}$$

770 where the last step follows again from Tonelli's theorem and the definition of $\tilde{\gamma}$. Combined with the
 771 previous chain of inequalities, we therefore have

$$\mathbb{E}_{\hat{\xi}_t} [(*)] \leq \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma}.$$

772 We can now plug back our bound in (12) to get

$$\begin{aligned} J_t(\mu_t, \nu) &\leq \inf_{(\text{proj}_S^{S \times A})_{\#} \lambda_t = \mu_t} \inf_{\tilde{\gamma} \in \Gamma(\lambda_t, \nu)} \int_{S \times A \times Z} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma} \\ &= \inf_{(\text{proj}_{S \times Z}^{S \times A \times Z})_{\#} \tilde{\gamma} \in \Gamma(\mu_t, \nu)} \int_{S \times A \times Z} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma} \\ &=: (\star). \end{aligned}$$

773 Now, we can use Lemma B.4 to disintegrate the optimizer in $(\star)$, which gives

$$(\star) = \inf_{\substack{\tilde{\gamma} \in \Gamma(\mu_t, \nu) \\ \alpha^{s,z} \in \Lambda(A)}} \int_{S \times Z} \int_A \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\alpha^{s,z} d\tilde{\gamma}$$

774 Since j_t, g_t, f_t are lower semianalytic and $\xi_t(\cdot | s, a)$ is a Borel stochastic kernel, j_t is lower semian-
 775 analytic by backward induction and so is the integrand. Therefore, we can apply Lemma B.5, which
 776 gives

$$(\star) = \inf_{\tilde{\gamma} \in \Gamma(\mu_t, \nu)} \int_{S \times Z} \inf_{a \in A} \mathbb{E}_w [g_t(s, a, w) + j_{t+1}(f_t(s, a, w), z)] d\tilde{\gamma}.$$

777 At this point, we recognize that the term in the integral coincides with the Bellman recursion for j_t :

$$j_t(s, z) = \inf_{a_t \in A} \mathbb{E}_{w_t} [g_t(s, a_t, w_t) + j_{t+1}(f_t(s, a_t, w_t), z)].$$

778 Hence,

$$J_t(\mu_t, \nu) \leq (\star) = \inf_{\tilde{\gamma} \in \Gamma(\mu_t, \nu)} \int_{S \times Z} j_t(s, z) d\tilde{\gamma}.$$

779 This concludes the inductive step and the proof of the theorem. $\square$

Two comments on our proof are in order. First, while the proof follows the inductive strategy of the proof of the deterministic separation in [25], there is a key difference: the next population state is random and depends on the realized empirical noise kernel $\hat{\xi}_t$. This requires various additional steps (and preliminary lemmas) in the proof. All these steps trivialize in the absence of noise and, thus, our proof recovers the deterministic separation in [25] when the dynamics are deterministic. Second, our result is an upper bound on the cost-to-go function. Intuitively, this conservatism originates from the following fact: The population state-action distribution can, in principle, choose future couplings *after* observing this realization, while our upper bound predicates that couplings are chosen *before* the noise is realized, averaging over the kernel $\xi_t(\cdot | s, a)$. This is precisely where Lemmas B.3 and B.6 yield inequalities rather than equalities. In the deterministic case, all these steps trivialize and we obtain equality as in [25].

We now prove Part II of the theorem.

Proof. For each timestep t , consider the function

$$\mathcal{K}[j_t](\mu_t, \nu) = \inf_{\gamma \in \Gamma(\mu_t, \nu)} \int_{S \times Z} j_t(s, z) d\gamma.$$

Let $\varepsilon > 0$. At each time t , we can construct a “separation-based” state-action distribution λ_t^ε as follows: given a population distribution μ_t , choose an $\frac{\varepsilon}{2T}$ -optimal coupling $\gamma_t^\varepsilon \in \Gamma(\mu_t, \nu)$ satisfying

$$\int_{S \times Z} j_t(s, z) d\gamma_t^\varepsilon \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{\varepsilon}{2T}, \quad (13)$$

and a universally measurable $\frac{\varepsilon}{2T}$ -optimal action selector $a_t^\varepsilon: S \times Z \rightarrow A$ satisfying, for all (s, z) ,

$$\mathbb{E}_w [g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z)] \leq j_t(s, z) + \frac{\varepsilon}{2T}. \quad (14)$$

Formally, we have

$$\lambda_t^\varepsilon := (\text{proj}_S^{S \times Z}, a_t^\varepsilon)_\# \gamma_t^\varepsilon.$$

In words, this state-action distribution assigns agents to targets through γ_t^ε and controls the individual agents through a_t^ε . Clearly, $\lambda_t^\varepsilon \in M(\mu_t)$. In the rest of the proof, we will show that λ_t^ε is ε -optimal, i.e.,

$$J_0^\varepsilon(\mu_0, \nu) := J_0^{\lambda_0^\varepsilon, \dots, \lambda_{T-1}^\varepsilon}(\mu_0, \nu) \leq \mathcal{K}[j_0](\mu_0, \nu) + \varepsilon. \quad (15)$$

Before doing so, we verify well-posedness of the all objects. The $\frac{\varepsilon}{2T}$ -optimal universally measurable selector a_t^ε exists by [29, Proposition 7.50]. For the existence of the $\frac{\varepsilon}{2T}$ -optimal coupling we work as follows. Since j_t is lower semianalytic (by backward induction on (2)), the map $\gamma \mapsto \int_{S \times Z} j_t d\gamma$ is lower semianalytic by [29, Corollary 7.48.1]. Further, the set $\{(\mu, \gamma) : \gamma \in \Gamma(\mu, \nu)\}$ is analytic. Indeed, fixing $\nu \in \mathcal{P}(W)$, define

$$\mathcal{G}_Z := \{(\mu, \gamma) \in \mathcal{P}(S) \times \mathcal{P}(S \times Z) : \gamma \in \Gamma(\mu, \nu)\}.$$

The maps $F_S(\gamma) := (\text{proj}_S^{S \times Z})_\# \gamma$ and $F_Z(\gamma) := (\text{proj}_Z^{S \times Z})_\# \gamma$ are continuous for the weak topologies, being the pushforward under a continuous map. Hence,

$$\mathcal{G}_Z = \{(\mu, \gamma) : F_S(\gamma) = \mu\} \cap \{(\mu, \gamma) : F_Z(\gamma) = \nu\}.$$

The first set is the preimage of the diagonal in $\mathcal{P}(S) \times \mathcal{P}(S)$ under the continuous map $(\mu, \gamma) \mapsto (\mu, F_S(\gamma))$, and the second is the preimage of the closed singleton $\{\nu\}$ under the continuous map $(\mu, \gamma) \mapsto F_Z(\gamma)$. Since these spaces are Polish, the diagonal and singleton are closed. By continuity F_S and F_Z , the set $\mathcal{G}_Z$ is closed, hence Borel and in particular analytic. Thus, we can apply [29, Proposition 7.50] to obtain the existence of a universally measurable map

$$\mu \mapsto \gamma_t^\varepsilon(\mu) \in \Gamma(\mu, \nu) \quad \text{with} \quad \int_{S \times Z} j_t d\gamma_t^\varepsilon(\mu) \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{\varepsilon}{2T}.$$

This shows the existence of a universally measurable selector $\gamma_t^\varepsilon(\mu)$. In particular, for fixed μ , $\gamma_t^\varepsilon(\mu)$ is a Borel probability measure and, being the pushforward under a universally measurable (which coincides $\gamma_t^\varepsilon(\mu)$ -a.e. with a Borel map [29, Lemma 7.27]) λ_t^ε is a well-defined Borel probability measure.

816 We now turn our attention to the desired bound (15). We now show by backward induction the more
817 general bound

$$J_t^\varepsilon(\mu_t, \nu) := J_t^{\lambda_t^\varepsilon, \dots, \lambda_{T-1}^\varepsilon}(\mu_t, \nu) \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{T-t}{T}\varepsilon, \quad (16)$$

818 which, evaluated at time $t = 0$, readily implies (15). The base case $t = T$ holds exactly by definition
819 of terminal cost and $j_T = g_T$:

$$J_T^\varepsilon(\mu_T, \nu) = \mathcal{K}[g_T](\mu_T, \nu) = \inf_{\gamma \in \Gamma(\mu_T, \nu)} \int_{S \times Z} g_T(s, z) d\gamma = \mathcal{K}[j_T](\mu_T, \nu).$$

820 Now fix $t \leq T$ and assume (16) holds at $t + 1$. By the induction hypothesis

$$J_t^\varepsilon(\mu_t, \nu) \leq \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_{S \times A \times W} g_t(s, a, w) d\rho_t + \mathcal{K}[j_{t+1}](\mu_{t+1}, \nu) \right] + \frac{T-t-1}{T}\varepsilon. \quad (17)$$

821 Next, we define

$$\bar{\gamma}_t^\varepsilon := (f_t, \text{Id}_Z)_\#(\sigma_\# \tilde{\gamma}_t^\varepsilon) \in \mathcal{P}(S \times Z)$$

822 where

$$\tilde{\gamma}_t^\varepsilon := (\text{Id}_{S \times Z}, a_t^\varepsilon)_\# \gamma_t^\varepsilon \otimes \hat{\xi}_t^{s, a_t^\varepsilon(s, z)} \in \mathcal{P}(S \times Z \times A \times W).$$

823 and we use the permutation consider the permutation

$$\sigma : S \times Z \times A \times W \rightarrow S \times A \times W \times Z, \quad \sigma(s, z, a, w) = (s, a, w, z),$$

824 to “re-order” the marginals. Informally, $\bar{\gamma}_t^\varepsilon$ is the law of $(f_t(s, a_t^\varepsilon(s, z), w), z)$ where $(s, z) \sim \gamma_t^\varepsilon$
825 and $w \sim \hat{\xi}_t^{s, a_t^\varepsilon(s, z)}$. We now show $\bar{\gamma}_t^\varepsilon \in \Gamma(\mu_{t+1}, \nu)$. Specifically, the second marginal is ν , since the
826 z -coordinate is unchanged and $(\text{proj}_Z^{S \times Z})_\# \gamma_t^\varepsilon = \nu$. The first marginal, instead, is the pushforward
827 of state-action-noise distribution ρ_t^ε associated with λ_t^ε under the dynamics f_t (i.e., the law of
828 $f_t(s, a_t^\varepsilon(s, z), w)$). This is precisely μ_{t+1} , which completes the proof of $\bar{\gamma}_t^\varepsilon \in \Gamma(\mu_{t+1}, \nu)$. By
829 definition of the infimum, we then obtain

$$\mathcal{K}[j_{t+1}](\mu_{t+1}, \nu) \leq \int_{S \times Z} j_{t+1}(s, z) d\bar{\gamma}_t^\varepsilon = \int_{S \times Z} \int_W j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z) d\hat{\xi}_t^{s, a_t^\varepsilon(s, z)} d\gamma_t^\varepsilon.$$

830 Taking expectation over noise realizations, using Fubini-Tonelli’s theorem, and applying Lemma A.9
831 we get

$$\begin{aligned} & \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_{S \times A \times W} g_t(s, a, w) d\rho_t + \mathcal{K}[j_{t+1}](\mu_{t+1}, \nu) \right] \\ & \leq \mathbb{E}_{\hat{\xi}_t^{s,a}} \left[\int_{S \times Z} \int_W g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z) d\hat{\xi}_t^{s, a_t^\varepsilon(s, z)} d\gamma_t^\varepsilon \right] \\ & = \int_{S \times Z} \mathbb{E}_w [g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z)] d\gamma_t^\varepsilon. \end{aligned}$$

832 Then, we use the $\frac{\varepsilon}{2T}$ -optimality of a_t^ε in (14) and of γ_t^ε in (13) to get

$$\begin{aligned} & \int_{S \times Z} \underbrace{\mathbb{E}_w [g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z)]}_{\leq j_t(s, z) + \varepsilon/(2T)} d\gamma_t^\varepsilon \\ & \leq \int_{S \times Z} j_t(s, z) d\gamma_t^\varepsilon + \frac{\varepsilon}{2T} \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{\varepsilon}{2T} + \frac{\varepsilon}{2T}. \end{aligned}$$

833 We can now combine this bound with (17) and get

$$J_t^\varepsilon(\mu_t, \nu) \leq \mathcal{K}[j_t](\mu_t, \nu) + \frac{\varepsilon}{T} + \frac{T-t-1}{T}\varepsilon = \mathcal{K}[j_t](\mu_t, \nu) + \frac{T-t}{T}\varepsilon.$$

834 This concludes the inductive step. The bound at $t = 0$ gives $J_0^\varepsilon(\mu_0, \nu) \leq \mathcal{K}[j_0](\mu_0, \nu) + \varepsilon$. $\square$

835 *Remark C.1* (Existence of optimal couplings under stronger regularity). The proof of Part 2 only
836 requires universally measurable ε -optimal coupling selectors, which exist under the lower semiana-
837 lytic assumptions used throughout Appendix B. Under the stronger assumption that the cost-to-go
838 functions $j_t : S \times Z \rightarrow \mathbb{R}_{\geq 0}$ are lower semicontinuous and the spaces are compact, the infimum
839 in $\mathcal{K}[j_t](\mu, \nu)$ is attained for each fixed (μ, ν) by the standard existence theorem for optimal cou-
840 plings [41, Theorem 4.1]. This pointwise existence result is distinct from the measurable-selection
841 argument needed above to construct a Markov policy depending on μ_t .

842 C.2 Proof of Corollary 3.3

843 *Proof.* We construct, for every $\varepsilon > 0$, an admissible separation policy whose cost is within $\frac{(T-t)\varepsilon}{T}$
 844 of $\mathcal{K}[j_t](\mu_t^N, \nu^N)$. Since $\varepsilon > 0$ is arbitrary, and J_t^N is the infimum of the expected cost over all
 845 admissible policies, the claim follows. We start by giving an ε -optimal separation policy $\pi_{\text{sep},t}^\varepsilon$ which
 846 attains the upper bound. We will then show that this policy admissible, we directly obtain the desired
 847 inequality.

848 To define $\pi_{\text{sep},t}^\varepsilon$ we proceed as follows. First, since μ_t^N and ν^N are both empirical measures over
 849 N atoms, the optimal transport problem $\mathcal{K}[j_t](\mu_t^N, \nu^N)$ is a finite-dimensional transportation linear
 850 program. As such, every coupling between μ_t^N and ν^N can be represented by a doubly-stochastic
 851 matrix $\Gamma = (\gamma_{ij})_{i,j=1}^N$ satisfying $\gamma_{ij} \geq 0$, $\sum_{i=1}^N \gamma_{ij} = \sum_{j=1}^N \gamma_{ij} = \frac{1}{N}$. By the Birkhoff-von
 852 Neumann theorem [42], every doubly stochastic matrix is a convex combination of permutation
 853 matrices; since the objective is linear, the minimum is attained by a scaled permutation matrix. Thus,

$$\mathcal{K}[j_t](\mu_t^N, \nu^N) = \frac{1}{N} \min_{\sigma_t \in S_N} \sum_{i=1}^N j_t(s_t^i, z^{\sigma_t(i)}),$$

854 where S_N denotes the group of permutations of $\{1, \dots, N\}$. Let σ_t^* attain this minimum and write
 855 $z_t^{*,i} := z^{\sigma_t^*(i)}$ for the target assigned to agent i . Given that agents are now assigned to targets, we
 856 can now consider the single-agent optimal control problem from s_t^i to the corresponding target
 857 $z_t^{*,i}$. Specifically, for $\varepsilon > 0$, there exists a universally measurable $\frac{\varepsilon}{T}$ -optimal action selector
 858 $a_t^\varepsilon: S \times Z \rightarrow A$, satisfying, for all $(s, z) \in S \times Z$:

$$\mathbb{E}_w [g_t(s, a_t^\varepsilon(s, z), w) + j_{t+1}(f_t(s, a_t^\varepsilon(s, z), w), z)] \leq j_t(s, z) + \frac{\varepsilon}{T}.$$

859 At this point, we define the separation policy $\pi_{\text{sep},t}^\varepsilon$ at time t . An agent at s_t^i executes the action
 860 $a_t^\varepsilon(s_t^i, z_t^{*,i})$. Informally, the optimal transport problem indicates the target, whereas the single-agent
 861 optimal control problem determines the optimal action to reach that target. Clearly, this policy is
 862 deterministic and Markovian, as it solely depends on $\mathbf{s}_t = (s_t^1, \dots, s_t^N)$. Measurability follows from
 863 the measurability arguments in the proof of Theorem 3.1.

864 At this point, it suffices to verify that this policy yields the bound on the cost. In particular, we show
 865 that the expected cost-to-go function under this policy, denoted by $J_t^{N,\varepsilon}(\mathbf{s}_t, \nu^N)$, is upper bounded
 866 as follows:

$$J_t^{N,\varepsilon}(\mathbf{s}_t, \nu^N) \leq \mathcal{K}[j_t](\mu_t^N, \nu^N) + \frac{T-t}{T}\varepsilon,$$

867 which gives the desired upper bound. We proceed by induction. The equality at time $t = T$ is trivial,
 868 as the terminal cost does not depend on the policy but only on the terminal state. Suppose then the
 869 statement holds at some time $t + 1$. Then,

$$\begin{aligned} J_t^{N,\varepsilon}(\mathbf{s}_t, \nu^N) &\leq \mathbb{E}_{w_t^i} \left[\frac{1}{N} \sum_{i=1}^N g_t(s_t^i, a_t^i, w_t^i) + \mathcal{K}[j_{t+1}](\mu_{t+1}^N, \nu^N) \right] + \frac{T-t-1}{T}\varepsilon \\ &= \frac{1}{N} \mathbb{E}_{w_t^i} \left[\sum_{i=1}^N g_t(s_t^i, a_t^i, w_t^i) + \min_{\sigma_{t+1} \in S_N} \sum_{i=1}^N j_{t+1}(f_t(s_t^i, a_t^i, w_t^i), z^{\sigma_{t+1}(i)}) \right] \\ &\quad + \frac{T-t-1}{T}\varepsilon \\ &\leq \frac{1}{N} \sum_{i=1}^N \mathbb{E}_{w_t^i} \left[g_t(s_t^i, a_t^i, w_t^i) + j_{t+1}(f_t(s_t^i, a_t^i, w_t^i), z^{\sigma_t^*(i)}) \right] + \frac{T-t-1}{T}\varepsilon \\ &\leq \frac{1}{N} \sum_{i=1}^N j_t(s_t^i, z^{\sigma_t^*(i)}) + \frac{\varepsilon}{T} + \frac{T-t-1}{T}\varepsilon \\ &= \frac{1}{N} \min_{\sigma \in S_N} \sum_{i=1}^N j_t(s_t^i, z^{\sigma(i)}) + \frac{T-t}{T}\varepsilon, \end{aligned}$$

where the first inequality is precisely the inductive assumption, the second inequality follows since σ_t^* is a potentially suboptimal permutation, last inequality follows from the use of the policy $\pi_{\text{sep},t}^\varepsilon$, and the last equality the definition of σ^* . This gives

$$J_t^{N,\varepsilon}(\mathbf{s}_t, \nu^N) \leq \mathcal{K}[j_t](\mu_t^N, \nu^N) + \frac{T-t}{T}\varepsilon,$$

which completes the induction.

Since $\varepsilon > 0$ is arbitrary, and J_t^N is the infimum of the expected cost over all admissible policies, letting $\varepsilon \downarrow 0$ gives

$$J_t^N(\mathbf{s}_t, \nu^N) \leq \mathcal{K}[j_t](\mu_t^N, \nu^N).$$

□

D Additional numerical results

This appendix provides complementary experimental results and full environment and hyperparameter specifications for the experiments in Section 5. Section D.1 covers the grid-world setup and additional metrics. Section D.2 details the scalability experiment. Section D.3 describes the Manhattan road network environment.

D.1 Additional details for the grid-world experiments in Section 5.1

Environment. The environment consists of an 11×30 grid with horizon $T = 100$ and $N = 16$ agents. At the beginning of each episode, agents are initialized uniformly at random in the first column, while the N targets are sampled uniformly in the last ten columns. The action space is $\mathcal{A} = \{\text{up}, \text{down}, \text{left}, \text{right}, \text{stay}\}$. At each timestep, the intended action is executed with probability $1 - p$ and replaced by a uniformly sampled action with probability p . We evaluate noise probabilities $p \in \{0, 0.1, 0.25\}$ under individual, local, and global noise. Individual noise samples this perturbation independently for each agent, local noise shares a perturbation among agents with the same state-action pair, and global noise shares one perturbation across the whole population. Their formal definitions are provided in Appendix A. After the transition, positions are clipped to remain inside the grid. In both MARL and the target-conditioned single-agent control problem in SALT, we introduce an induced bias to facilitate learning. For the MARL baselines, an agent becomes inactive when it first reaches an unoccupied target. For SALT, learning is target-conditioned and single-agent: the agent is associated with one assigned target z , and becomes inactive when it reaches that assigned target (note, in particular, that this is not coupling agents, but rather facilitates the learning of the single-agent target-conditioned problem). Absorbed agents incur zero further stage cost, ignore future actions and perturbations, and remain fixed for the rest of the episode.

Reward structure. Each active agent incurs a stage cost of 1 at every timestep. At the horizon, an additional optimal transport cost $K[c](\mu_T, \nu)$ is applied, where the cost function c is the Manhattan distance between grid cells. The combined objective therefore penalizes both travel time and terminal coverage quality.

Baselines. We compare against MAPPO and IPPO, VDN, and QMIX. All baselines use the same episode budget and random seeds. All neural policies use two hidden layers with 64 units and tanh activations. The separation-based neural methods are trained as goal-conditioned single-agent controllers. For SALT-PPO, SALT-A2C, and SALT-DDQN, the policy input is

$$\left(\frac{r}{H_g}, \frac{c}{W_g}, \frac{r_z}{H_g}, \frac{c_z}{W_g}, \frac{t}{T-1}, \frac{\|s - z\|_1}{H_g + W_g} \right),$$

where (r, c) is the current agent position, (r_z, c_z) is the currently assigned target, t is the timestep, h and w are the grid height and width, and $H_g = h - 1$, $W_g = w - 1$ are the maximum row and column indices. Thus, the separation policy never observes the full population during action selection; coordination of the agents enters only through the optimal transport assignment that chooses the target z .

912 MAPPO, IPPO, QMIX, and VDN receive target-relative local observations. For each agent i , the
 913 local observation has dimension $5 + 3M$, where $M = N$ is the number of targets:

$$\left(\frac{r_i}{H_g}, \frac{c_i}{W_g}, \frac{i}{N-1}, \frac{t}{T-1}, a_i, \frac{\Delta r_{i1}}{H_g}, \frac{\Delta c_{i1}}{W_g}, b_1, \dots, \frac{\Delta r_{iM}}{H_g}, \frac{\Delta c_{iM}}{W_g}, b_M \right).$$

914 Here a_i is an active-agent indicator, $\Delta r_{ij} = r_{z_j} - r_i$ and $\Delta c_{ij} = c_{z_j} - c_i$ are the relative coordinates
 915 of target j from agent i , and b_j indicates whether target j remains unoccupied. The target set is
 916 therefore visible to each baseline agent as an ordered list of relative target offsets and target-availability
 917 flags. MAPPO additionally uses a centralized critic state of dimension $3N + 1 + 3M$, containing all
 918 normalized agent positions, all agent active flags, the timestep, all normalized target positions, and all
 919 target-availability flags. QMIX uses the same global state inside its monotone mixing network with
 920 mixer embedding dimension 32. IPPO and VDN do not use a separate global critic state; IPPO uses
 921 a local critic, and VDN uses additive value decomposition. All neural methods are optimized with
 922 Adam.

Hyperparameters We summarize in Tables 2, 3, and 4 the used hyperparameters.

Table 2: Shared grid-world training budget and evaluation settings for Section 5.1.

Quantity	Value
Grid / horizon	$11 \times 30, T = 100$
Agents / targets	$N = 16$
Training seed	0
Evaluation seeds	20 paired seeds
Noise probabilities	0, 0.1, 0.25
Noise modes	individual, local, global
Training episodes	320000
Environment interactions	$320000 \times 16 \times 100 = 5.12e8$
Batch episodes	64
Update batches for neural methods	5000
Step cost	1
Evaluation target region	columns 20–29
Evaluation rematching for SALT	every timestep

923

Table 3: Hyperparameters used for SALT in Section 5.1.

Hyperparameter	SALT-Q	SALT-DDQN	SALT-PPO	SALT-A2C
Discount γ	0.98	0.98	0.98	0.98
Learning rate	$0.25 \rightarrow 0.02$	$3 \cdot 10^{-4}$	actor $3 \cdot 10^{-4}$ critic 10^{-3}	actor $3 \cdot 10^{-4}$ critic 10^{-3}
Hidden size	–	64	64	64
GAE λ	–	–	0.95	0.95
PPO clip / epochs	–	–	0.2 / 4	–
Entropy coefficient	–	–	0.01	0.01
Max grad norm	–	0.5	0.5	0.5
Exploration α	$0.5 \rightarrow 0.05$	$0.5 \rightarrow 0.05$	$0.5 \rightarrow 0.05$	$0.5 \rightarrow 0.05$
Exploration decay (interactions)	$1.12 \cdot 10^8$	$1.12 \cdot 10^8$	$1.12 \cdot 10^8$	$1.12 \cdot 10^8$
Replay capacity	–	500000	–	–
Replay warmup / minibatch	–	5000 / 256	–	–
Learner steps per batch	–	64	–	–
Target update period	–	500 learner steps	–	–

924 **Additional performance metrics** Finally, we here complement the average travel time results
 925 reported in Figure 2. Since the task requires both successful navigation and coordination among
 926 agents, we evaluate the algorithms using several complementary metrics. The average travel time of
 927 the population in Figure 2 measured *how quickly* the agents covered the target distribution. Here, we

Table 4: Hyperparameters used for MARL baselines in Section D.1.

Hyperparameter	MAPPO	IPPO	QMIX	VDN
Discount γ	0.98	0.98	0.98	0.98
Learning rate	actor $3 \cdot 10^{-4}$ critic 10^{-3}	actor $3 \cdot 10^{-4}$ critic 10^{-3}	$3 \cdot 10^{-4}$	$3 \cdot 10^{-4}$
Hidden size	64	64	64	64
GAE λ	0.95	0.95	–	–
PPO clip / epochs	0.2 / 4	0.2 / 4	–	–
Entropy coefficient	0.01 $\rightarrow$ 0.001	0.01 $\rightarrow$ 0.001	–	–
Max grad norm	0.5	0.5	0.5	0.5
Actor minibatch	4096	4096	–	–
Critic minibatch	1024	4096	–	–
Replay capacity	–	–	100000	100000
Minimum replay size	–	–	2048	2048
Replay batch size	–	–	512	512
Updates per batch	–	–	1	1
Target update period	–	–	50 batches	50 batches
Exploration α	–	–	0.5 $\rightarrow$ 0.05	0.5 $\rightarrow$ 0.05
Exploration decay	–	–	160000 episodes	160000 episodes
QMIX mixer embedding	–	–	32	–

	Individual noise			Local noise			Global noise		
SALT-Q	0.65	0.65	0.50	0.65	0.70	0.00	0.65	0.50	0.50
SALT-DDQN	0.55	0.15	0.15	0.55	0.00	0.60	0.55	1.90	1.90
SALT-PPO	0.95	0.55	1.70	0.95	0.15	1.40	0.95	1.35	2.05
SALT-A2C	0.65	0.80	1.20	0.65	0.00	0.20	0.65	0.50	0.15
MAPPO	12.75	12.10	14.90	12.75	12.95	14.65	12.75	13.30	14.85
IPPO	10.35	14.70	20.05	10.35	9.65	12.40	10.35	14.00	14.05
QMIX	20.30	23.55	24.75	20.30	21.60	30.30	20.30	22.45	21.95
VDN	20.70	31.35	25.55	20.70	35.90	28.25	20.70	22.05	22.75
	0	0.10	0.25	0	0.10	0.25	0	0.10	0.25
	Noise probability			Noise probability			Noise probability		

Figure 4: Optimal transport cost between the final agent positions and the set of targets against environment noise for our separation principle.

928 focus on understanding *how well* and *how successfully* they cover it. Figure 4 gives a measure of
 929 coverage quality by reporting the empirical optimal transport cost between the final agent positions
 930 and the set of targets. This metric measures the minimum average Manhattan distance required to
 931 match each agent to a target. Let

$$\mu_T^N = \frac{1}{N} \sum_{i=1}^N \delta_{s_T^i}, \quad \nu^N = \frac{1}{N} \sum_{j=1}^N \delta_{z^j}.$$

932 We report the empirical terminal optimal transport cost

$$\mathcal{K}[d](\mu_T^N, \nu^N) = \min_{\sigma \in S_N} \frac{1}{N} \sum_{i=1}^N d(s_T^i, z^{\sigma(i)}),$$

933 where S_N is the set of permutations of $\{1, \dots, N\}$, s_T^i denotes the final position of agent i , z^j denotes
 934 the target locations, $d(\cdot, \cdot)$ is the Manhattan distance, and σ is the optimal assignment between agents
 935 and targets. Hence, a value of zero indicates that every target is perfectly occupied by an agent. This
 936 allows us to distinguish between different agents failures: populations that almost reached all targets
 937 should be valued more positively than populations that just moved randomly.

938 Finally, Figure 5 focuses on the agents' ability to successfully navigate the goals within the episode
 939 horizon. Specifically, the reach rate quantifies the fraction of agents that reach a target during the

	Individual noise			Local noise			Global noise		
SALT-Q	100	99.3	100	100	99.3	100	100	100	99.3
SALT-DDQN	100	100	100	100	100	100	100	100	100
SALT-PPO	100	100	100	100	100	100	100	100	100
SALT-A2C	100	100	100	100	100	100	100	100	100
MAPPO	77.1	76.4	70.7	77.1	78.6	73.6	77.1	75.7	75.7
IPPO	85.7	78.6	71.4	85.7	81.4	75	85.7	80	79.3
QMIX	56.4	52.9	58.6	56.4	48.6	42.1	56.4	56.4	56.4
VDN	55	33.6	52.1	55	30.7	38.6	55	50	51.4
	0	0.10	0.25	0	0.10	0.25	0	0.10	0.25
	Noise probability			Noise probability			Noise probability		

Figure 5: Reach rate against environment noise. Entries are means and darker cells indicate better relative performance within this metric.

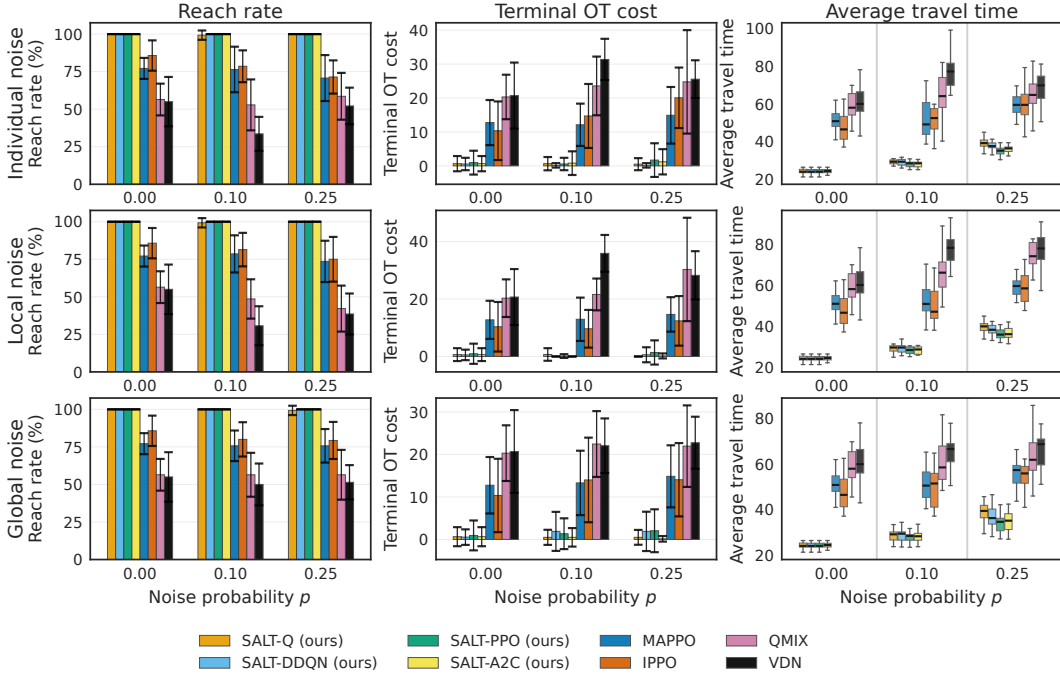


Figure 6: Combined reach rate, terminal optimal transport cost and average travel time.

episode. Figure 6 summarizes the three evaluation metrics used in Figures 2, 5 and 4: average travel time, terminal optimal transport cost, and reach rate. Each value is averaged over the same paired evaluation seeds for a fixed algorithm, noise model, and noise probability p . The metrics are reported on their native scales: higher is better for reach rate, while lower is better for terminal optimal transport cost and average travel time. Across all three metrics, the SALT variants achieve higher reach rates and lower terminal optimal transport costs and travel times than the MARL baselines.

D.2 Additional details for the scalability experiment

To study sample complexity, we consider square grids. All N agents are initialized in the left-bottom corner, and targets are uniformly sampled in the whole grid. To isolate population-size scaling from task difficulty, we scale grid dimensions proportionally with N so that target density remains approximately constant (see Table 5). All experiments are conducted under the individual-noise model with noise probability $p = 0.1$. We instantiate SALT with PPO as the single-agent learner and compare it against MAPPO. The hyperparameters for both methods follow Table 3 and Table 4 above.

Table 5: Population sizes and grid sizes used in the scalability experiment.

Agents N	2	4	6	8	16	32	64	128
Grid size	10^2	14^2	17^2	20^2	28^2	40^2	57^2	80^2
Horizon	20	28	34	40	56	80	114	160
Planned batches	781250	279018	153186	97656	34877	12207	4283	1526

The batch size is 64 episodes; the number of batches is chosen separately for each population size so that the interaction budget is approximately constant and equal to 2×10^9 .

D.3 Towards a realistic, time-varying urban logistics task

Manhattan environment and dynamics. The Manhattan environment is a directed road graph extracted from OpenStreetMap [36] for southern Manhattan. OpenStreetMap data are made available under the Open Data Commons Open Database License (ODbL). In the current training run, the processed graph contains $|V| = 2088$ nodes. Each node corresponds to an intersection and each directed edge to an admissible road segment. Nominal edge travel times are obtained from road length and road-type-dependent speeds.

Traffic signals are modeled at the edge level. Each edge has a traffic-light cycle of 90 seconds. Green-phase durations depend on road type: primary roads are green for 50% of the cycle, secondary and residential roads for 30%, and motorway, trunk, tertiary, and unclassified roads are always green. Each signal receives a randomized phase offset. When an agent traverses edge e at time t , the realized transition time is the travel time plus the signal waiting time induced by the edge’s phase.

We additionally inject stochastic congestion noise on selected road types. For a traversed edge e , the travel time is multiplied by

$$1 + u_t, \quad u_t \sim \text{Unif}(0, \eta_e),$$

where $\eta_e = 0.1$ for primary roads and $\eta_e = 0.15$ for secondary roads and u_t is independently drawn across edges.

Time discretization and Q-table. The Manhattan simulator evolves in event time: one decision step corresponds to traversing one outgoing road segment, not to advancing by a fixed one-second tick. The tabular Q-learner uses a discrete time coordinate with resolution

$$\Delta t = 1 \text{ second.}$$

Since all traffic lights use a 90 second cycle, the Q-table contains $T_{\text{disc}} = 90/\Delta t = 90$ time slices. The stored time coordinate is the traffic phase, not the unbounded absolute clock time. Thus, for physical time t and cycle period $P = 90$ seconds, the table index is $\tau(t) = (\text{round}(t/\Delta t)) \bmod T_{\text{disc}}$.

The Q-table is target-conditioned and indexed by current node v , pickup node z , traffic phase τ , and outgoing action a :

$$Q(v, z, \tau, a).$$

In the Manhattan graph this gives shape

$$|V| \times |V| \times T_{\text{disc}} \times \max_v |\text{out}(v)| = 2088 \times 2088 \times 90 \times 4,$$

where $|\text{out}(v)|$ is the maximum outgoing degree of the graph. Invalid padded actions are masked during action selection and Bellman backups. The table is stored in `bfloat16` to reduce memory usage.

For consistency with the tabular state space, edge travel times are rounded to the same grid before training. If $T_{v,a}$ is the continuous travel time of edge (v, a) , the learner uses $\hat{T}_{v,a} = \Delta t \text{round}(T_{v,a}/\Delta t)$. After traversing the edge, the simulator computes the traffic-light waiting time from the resulting arrival phase. For edge period $P_{v,a}$, offset $o_{v,a}$, and green duration $g_{v,a}$,

$$\phi = (t + \hat{T}_{v,a} + o_{v,a}) \bmod P_{v,a},$$

988 and

$$W_{v,a}(t) = \begin{cases} 0, & \phi < g_{v,a}, \\ P_{v,a} - \phi, & \phi \geq g_{v,a}. \end{cases}$$

989 The next physical time is

$$t' = t + \hat{T}_{v,a} + W_{v,a}(t),$$

990 and the next stored time is rounded back to the discrete grid and reduced modulo the traffic-light
991 period:

$$\tilde{t}' = (\Delta t \text{ round}(t'/\Delta t)) \bmod P_{v,a}.$$

992 This $\tilde{t}'$ is the time component used for the next Q-table lookup. The reward uses the realized delay
993 $\hat{T}_{v,a} + W_{v,a}(t)$, scaled in minutes, plus the pickup bonus when the target is reached.

994 **Task setup.** Starts are sampled from three fixed depot nodes. These depots are selected once before
995 training using betweenness centrality, with a preference for nodes in different taxi zones when zone
996 metadata is available. Pickup locations are sampled uniformly from all graph nodes.

997 Before routing, pickups are assigned to agents by solving a linear assignment problem with the
998 Hungarian algorithm. Since starts and pickups have equal mass, this is the empirical optimal-
999 transport problem. The assignment cost is the negative current target-conditioned Q-value estimate,
1000 used as a proxy for the single-agent cost-to-go function j_t .

1001 **Learning algorithm.** The Manhattan experiment uses tabular Q-learning. The Q-learning update
1002 uses learning rate 0.1 and discount factor $\gamma = 1.0$. Exploration is ε -greedy, with

$$\varepsilon_{\text{start}} = 1.0, \quad \varepsilon_{\text{end}} = 0.01.$$

1003 The decay is linear over 90% of the total training steps. The training budget is 5,000,000 episodes.

1004 The Q-table is initialized from offline shortest-path travel-time estimates (hence blind to noise and
1005 time-varying aspects) before learning. In the current configuration, initialization is applied to the
1006 initial time slice only, rather than all 90 time slices, to avoid the large memory and compilation cost
1007 of full time-slice initialization.

1008 **Reward.** At each step, an active agent receives the negative realized delay in minutes plus a pickup
1009 bonus upon reaching the target:

$$r_t = -\frac{\text{travel}_t + \text{wait}_t}{60} + 5 \cdot \mathbf{1}\{\text{pickup reached}\}.$$

1010 Completed agents enter an absorbing state and receive zero reward thereafter.

1011 **Evaluation and logging.** Training metrics are logged every 1,000 episodes. Evaluation is performed
1012 every 10,000 episodes. Final evaluation samples 500 sets of starts and pickups and evaluates all set
1013 combinations using Hungarian matching. Experiments are run with JAX [43].

1014 D.4 Compute resources

1015 All reported experiments were run on a single-GPU workstation. Unless otherwise stated, each run
1016 used one NVIDIA GeForce RTX 4090 GPU with 24 GB of GPU memory (24564 MiB as reported by
1017 the system) and 450 W maximum board power. The machine used 20 physical CPU cores and 28
1018 logical CPUs. The experiments were run on Linux 6.8.0-90 with CPython 3.10.12.

1019 For the Manhattan road-network experiments, a representative end-to-end run used four parallel
1020 workers and ran for 46.26 GPU-hours. All grid-world experiments were run on Apple M3 Max
1021 MacBook Pro, 14 CPU cores. The average training job took 40.3 minutes, with an average of 13.2
1022 minutes for our first experiment (performance) and 2.70 hours for the second experiment (scalability).
1023 A full experimental condition, including all algorithms trained for that setting, took 3.47 hours on
1024 average (1.76 hours for performance and 5.40 hours for scalability).

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